

V:\Sacramento Co-SA-13108-Alta Mesa Rd Bridge Replacement\CADD\Structures\02 AM_IP.dwg 10/29/2020 1:25 PM USERNAME => \$USER DATE PLOTTED => \$DATE TIME PLOTTED => \$TIME

GENERAL NOTES
LOAD AND RESISTANCE FACTOR DESIGN

DESIGN:
AASHTO LRFD Bridge Design Specifications, 8th edition with California Amendments

SEISMIC DESIGN:
Caltrans Seismic Design Criteria (SDC), Version 2.0 dated October 2017 April 2019

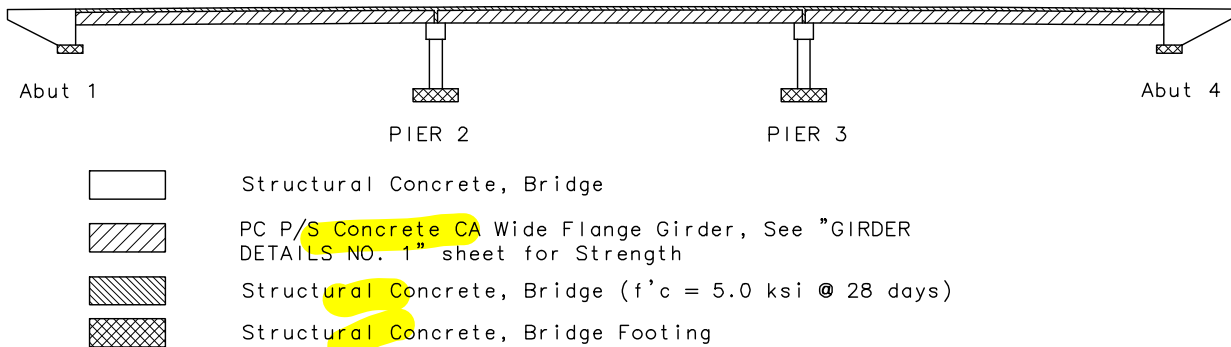
DEAD LOAD:
Includes 35 psf for future wearing surface. The deck load between the girders has been increased by a factor of 10% to allow for the use of steel deck forms.

LIVE LOADING:
HL93 and permit design load.

SEISMIC LOADING:
Soil Profile: Vs30 = 360 m/s (1200 ft/s)
Moment Magnitude: 6.8
Peak Ground Acceleration 0.23g
See ARS Curve

CONCRETE:
fy = 60 ksi
f'c = 3.6 ksi unless otherwise noted
n = 8
See "PRESTRESSING NOTES" on "GIRDER DETAILS NO. 1" sheet

PILE:
See "PILE DATA TABLE" on "FOUNDATION PLAN" sheet.



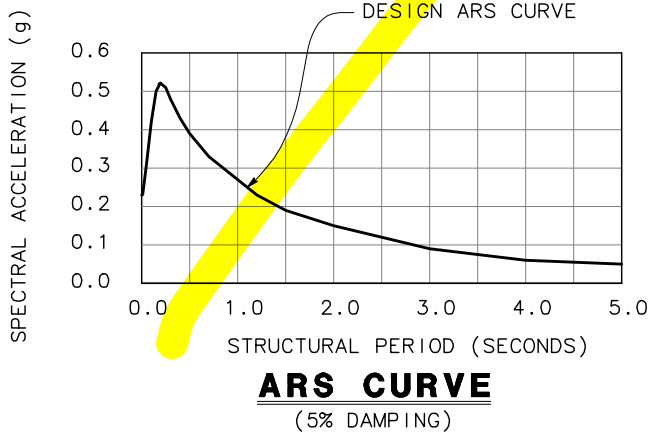
CONCRETE STRENGTH AND TYPE LIMITS
NO SCALE

QUANTITIES

BRIDGE REMOVAL	X	LS
STRUCTURE EXCAVATION (BRIDGE)	X	CY
STRUCTURE BACKFILL (BRIDGE)	X	CY
FURNISH PILING (CLASS 200) (ALTERNATIVE X)	X	LF
DRIVE PILE (CLASS 200) (ALTERNATIVE X)	X	EA
STRUCTURAL CONCRETE, BRIDGE FOOTING	X	CY
STRUCTURAL CONCRETE, BRIDGE	X	CY
STRUCTURAL CONCRETE, APPROACH SLAB (TYPE EQ)	X	CY
FURNISH PRECAST PRESTRESSED CONCRETE BULB-TEE GIRDER (115')	X	EA
ERECT PRECAST PRESTRESSED CONCRETE GIRDER	X	EA
JOINT SEAL (MR 1 1/2")	X	LF
BAR REINFORCING STEEL (BRIDGE)	X	LB
HEADED BAR REINFORCEMENT	X	EA
CALIFORNIA ST-75 BRIDGE RAIL	X	LF
MISCELLANEOUS METAL (BRIDGE)	X	LB

STANDARD PLANS DATED 2018

A3A-A3C	ABBREVIATIONS
A10A-A10E	LINES AND SYMBOLS
A10F-A10G	LEGEND - SOIL
A78F1	DOUBLE THRIE BEAM BARRIER - CONNECTIONS TO BRIDGE RAILINGS WITHOUT SIDEWALKS
A78F2	SINGLE THRIE BEAM BARRIER - CONNECTIONS TO BRIDGE RAILINGS WITHOUT SIDEWALKS
A62C	LIMITS OF PAYMENT FOR EXCAVATION AND BACKFILL - BRIDGE
RSP B0-1	BRIDGE DETAILS
B0-3	BRIDGE DETAILS
B0-5	BRIDGE DETAILS
B0-13	BRIDGE DETAILS
RSP B2-8	PILE DETAILS CLASS 200
B3-3A	RETAINING WALL TYPE 1A (CASE 1)
B6-21	JOINT SEALS (MAXIMUM MOVEMENT RATING = 2")
B9-4	STRUCTURE APPROACH TYPE EQ(10)
B9-5	STRUCTURE APPROACH SLAB DETAILS
B9-6	DRAINAGE DETAILS
T3A	TEMPORARY RAILING (TYPE K)



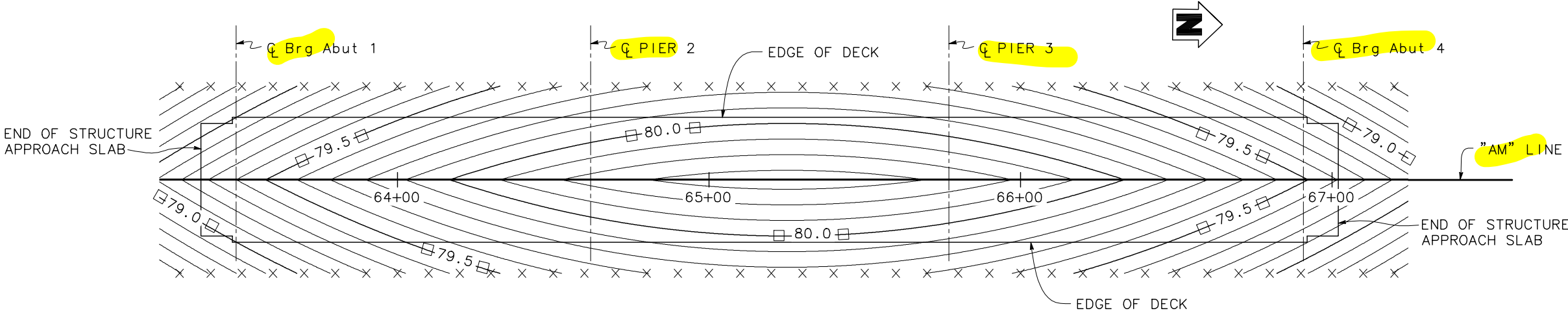
INDEX TO PLANS

SHEET NO.	TITLE
1	GENERAL PLAN
2	INDEX TO PLAN
3	DECK CONTOURS
4	FOUNDATION PLAN
5	ABUTMENT 1 LAYOUT
6	ABUTMENT 4 LAYOUT
7	ABUTMENT DETAILS No. 1
8	ABUTMENT DETAILS No. 2
9	PIER LAYOUT
10	PIER DETAILS
11	TYPICAL SECTION
12	GIRDER LAYOUT
13	GIRDER DETAILS NO. 1
14	GIRDER DETAILS NO. 2
15	GIRDER DETAILS NO. 3
16	GIRDER DETAILS NO. 4
17	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.1
18	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.2
19	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.3
20	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.4
21	ROCK SLOPE PROTECTION
22	LOG OF TEST BORINGS (1 OF 2)
23	LOG OF TEST BORINGS (2 OF 2)

DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)	
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM			POST MILES	INDEX TO PLAN	
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE	ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)	SHEET 2 OF 23
				FILE => \$REQUEST	CONTRACT NO.: X PROJECT ID:		

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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			11/6/20 DATE		
PLANS APPROVAL DATE			11/6/20		
The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.			REGISTERED PROFESSIONAL ENGINEER No. 81809 Exp. 3/31/22 CIVIL STATE OF CALIFORNIA		
SACRAMENTO COUNTY 411 BRANCH CENTER ROAD SACRAMENTO, CA 95827					
MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					

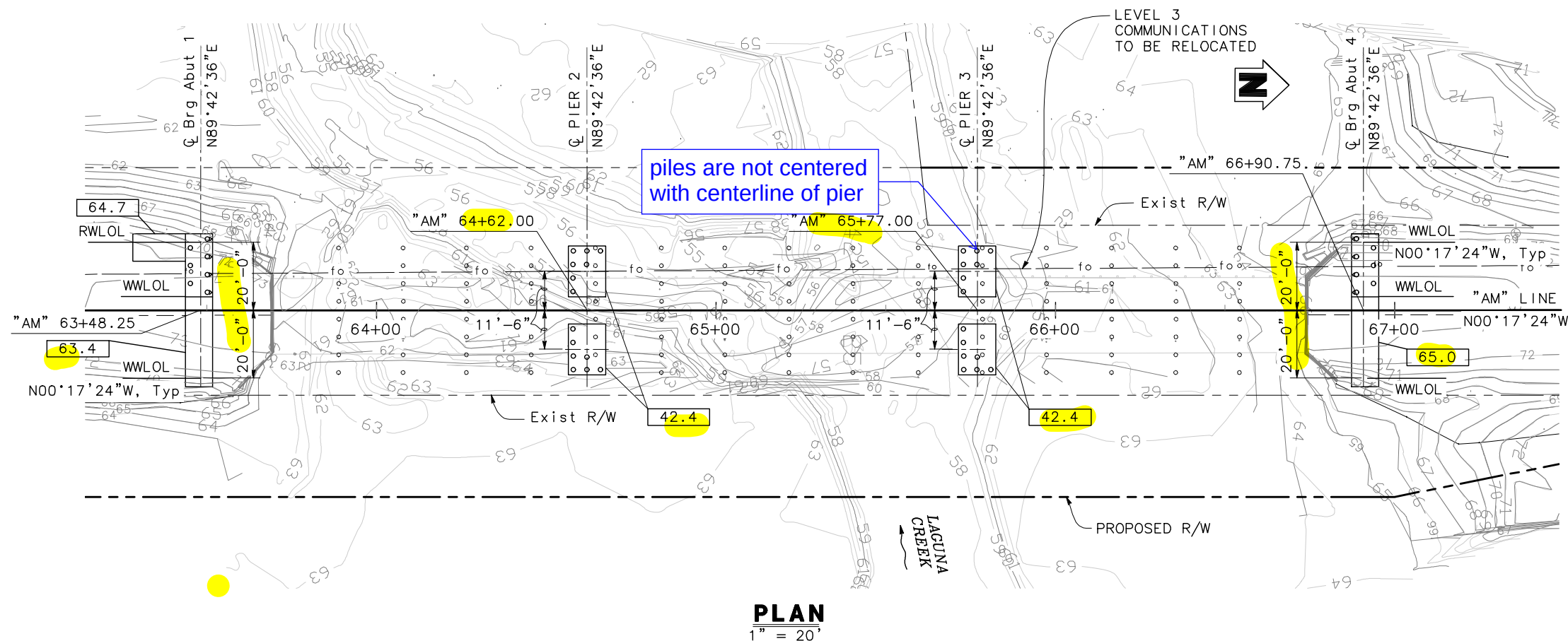


- NOTES:
1. Contour interval is 0.1 feet.
 2. Contours do not include camber settlement.
 3. X - Indicates 10 foot intervals.
 4. □ - Indicates half foot contour.

DECK CONTOURS
1" = 20'

DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)	
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				FILE => \$REQUEST	CONTRACT NO.: X		PROJECT ID:

DATE PLOTTED => \$DATE USERNAME => \$USER TIME PLOTTED => \$TIME



DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No.	TOTAL SHEETS
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 11/6/20
 REGISTERED STRUCTURAL ENGINEER DATE

PLANS APPROVAL DATE _____

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REGISTERED PROFESSIONAL ENGINEER
 TRUONG VIET PHAM
 No. 81809
 Exp. 3/31/22
 CIVIL
 STATE OF CALIFORNIA

SACRAMENTO COUNTY
 411 BRANCH CENTER ROAD
 SACRAMENTO, CA 95827

MARK THOMAS & COMPANY, INC.
 7300 FOLSOM BOULEVARD, SUITE 203
 SACRAMENTO, CA 95826

BENCHMARK:

Bronze tablet set in concrete stamped "U.S.C & G.S. B.M. U-953, 1962" located 22 feet East of center Alta Messa Rd and 12 feet South of the extended centerline of Langford Rd, and 3 feet East of a guy pole, 4 feet West of fence, 1.5 feet North of metal witness post and 0.45 Mi North of Laguna Creek.

Elevation 78.86 feet (NAVD88)

LEGEND:

- 
 Indicates Bottom of Footing Elevation (feet)
 ○ OR ○ Indicates Driven Pile (all piles not shown)
 ° Indicates Exist Pile Extension

PILE DATA TABLE

Location	Pile Type	Nominal Resistance (kips)		Design Tip Elevations (ft)	Specified Tip Elevations (ft)	Nominal Driving Resistance (kips)
		Compression	Tension			
Abut 1	CLASS 200 Alt "W"	360	N/A	22.0(a), 37.0(d)	22.0	360
PIER 2	CLASS 200 Alt "W"	310	120	1.0(a), 9.0(b), 16.0(d)	1.0	310
PIER 3	CLASS 200 Alt "W"	310	120	1.0(a), 9.0(b), 16.0(d)	1.0	310
Abut 4	CLASS 200 Alt "W"	360	N/A	22.0(a), 38.0(d)	22.0	360

NOTES:

1. Design tip elevations are controlled by (a) Compression, (b) **Tension**
2. Do not raise specified tip elevation.

NOTE :

THE CONTRACTOR SHALL VERIFY ALL
CONTROLLING FIELD DIMENSIONS BEFORE
ORDERING OR FABRICATING ANY MATERIAL.

SCOUR DATA TABLE

Location	Long Term (Degradation and Contraction) Scour Elevation (ft)	Short Term (Local) Scour Depth (ft)
Abut 1	69.0	5.6
PIER 2	51.0	7.7
PIER 3	51.0	8.6
Abut 4	69.0	3.6

HYDROLOGIC SUMMARY

Drainage Area 114.9 Square Mile

	Design Flood	Base Flood	Overtopping Flood
Frequency (Years)	50	100	N/A
Discharge (Cubic Foot per Sec)	15,750	18,260	N/A
Water Surface (Elevation at Bridge)	70.2	70.6	N/A

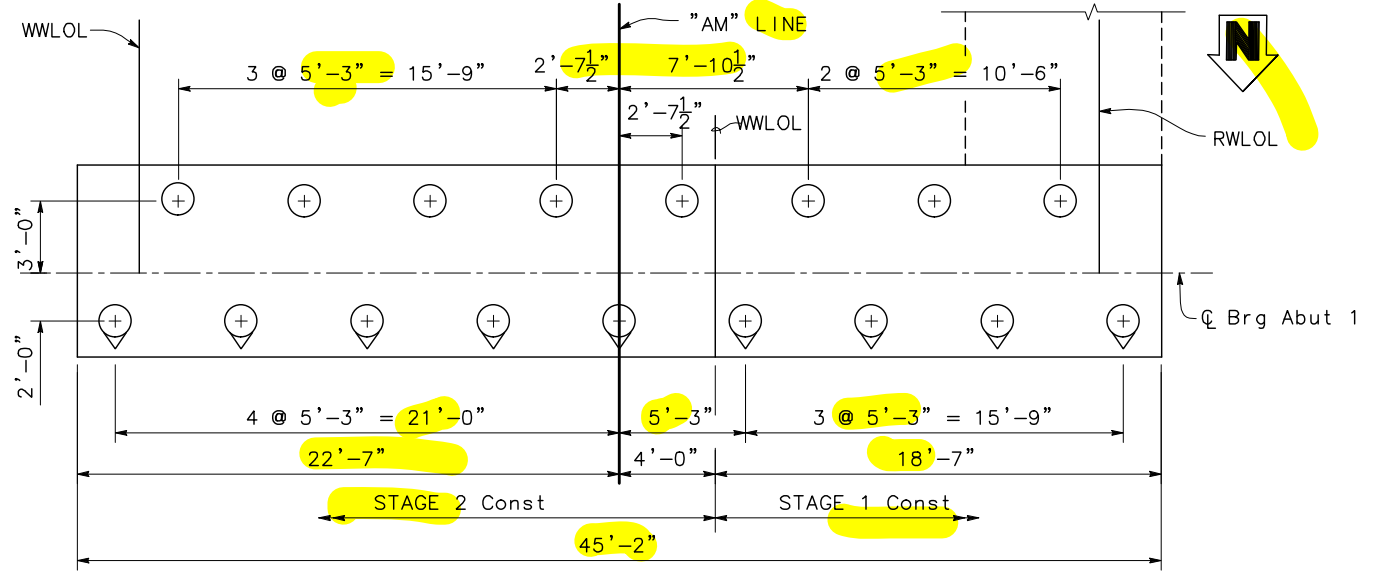
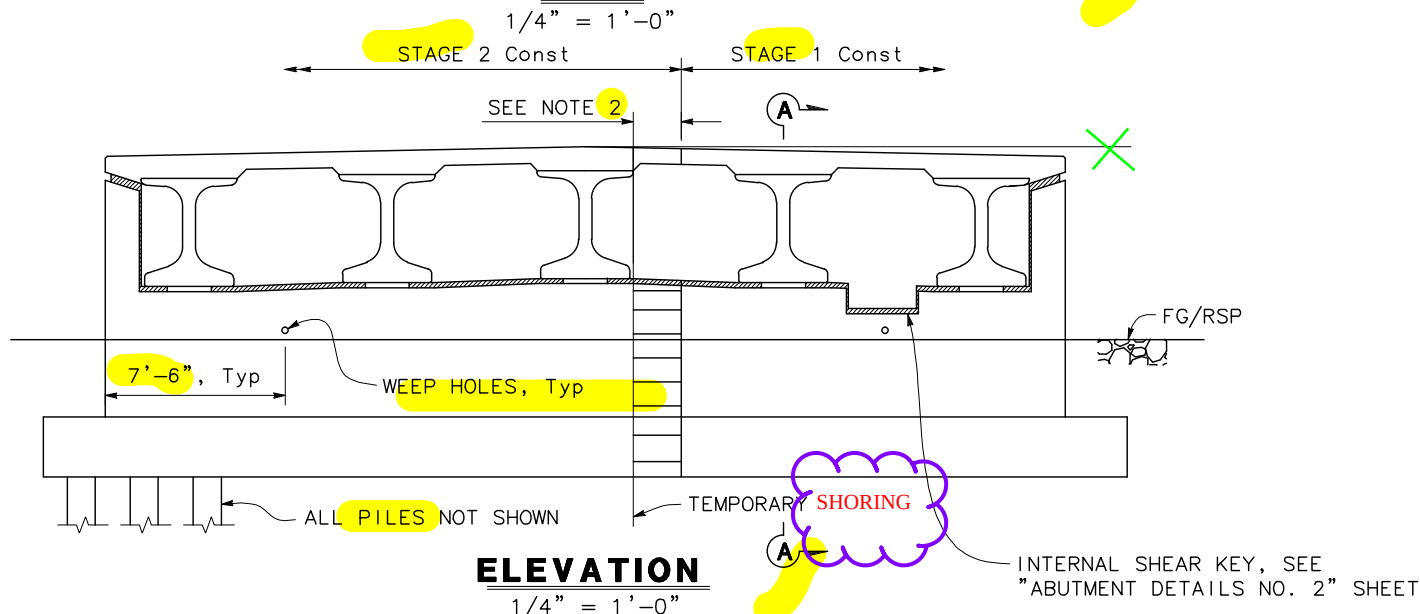
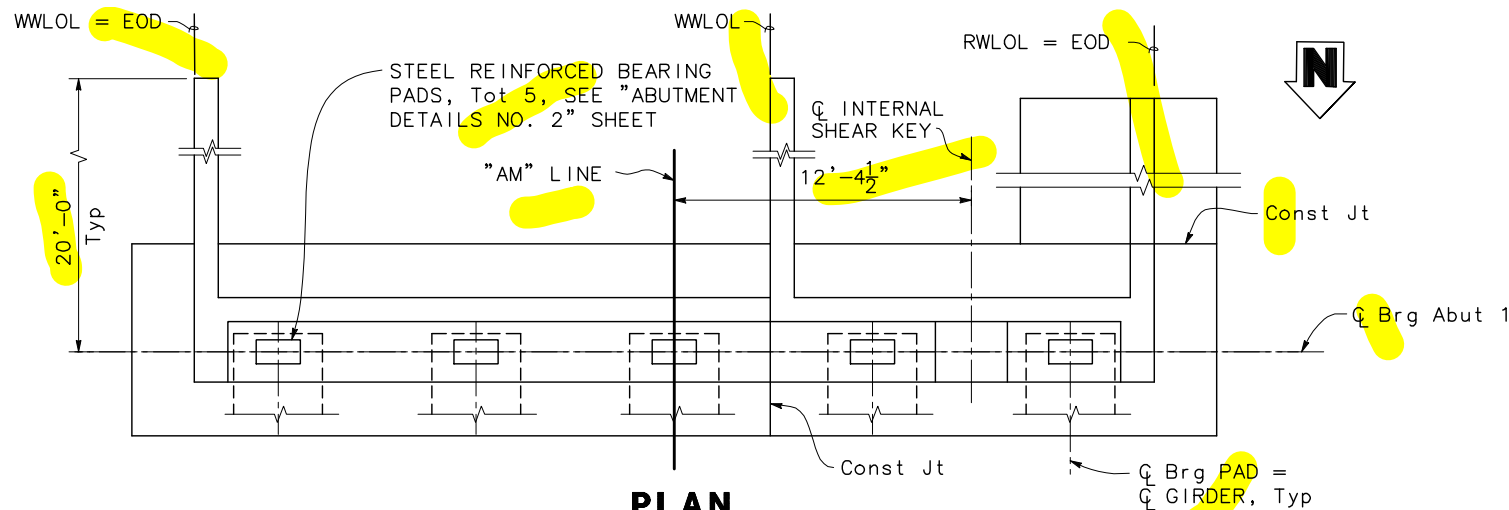
Flood plain data are based upon information available when the plans were prepared and are shown to meet federal requirements. The accuracy of said information is not warranted by the County and interested or affected parties should make their own investigation.

Hydrologic/ Hydraulic Study Elevation was based on NAVD88.

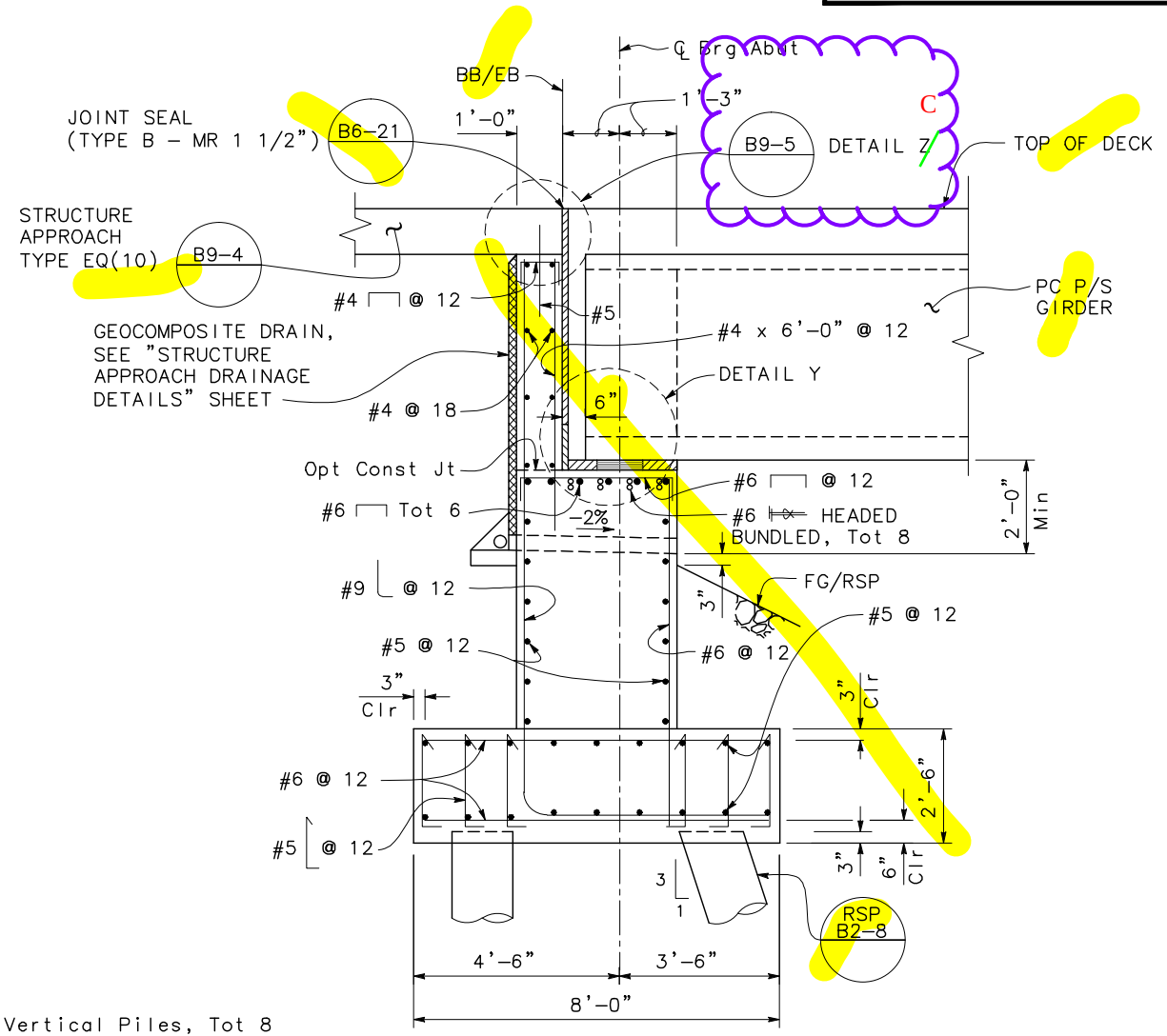
95% SUBMITTAL

		SCALE: No Scale		VERT.DATUM: NGVD88		HORIZ.DATUM:		DESIGN BY T. PHAM		CHECKED Y. REN		PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION				JASON HICKEY PROJECT ENGINEER		BRIDGE NO. 24C0606		LAGUNA CREEK BRIDGE (REPLACE)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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NOTE:
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- LEGEND:
- (+) Indicates Vertical Piles, Tot 8
 - (+) Indicates Battered Piles, Tot 9

- NOTES:
- For "DETAIL Y" AND "DETAIL Z", see "ABUTMENT DETAILS NO. 2" sheet.
 - Extend footing and abutment reinforcement 2'-0" beyond construction joint to lap with stage 2 construction. Reinforcement may be bent as necessary.
 - For Abut 1 West Retaining Wall details, see "ABUTMENT DETAILS NO. 1" sheet.

DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
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MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					

REGISTERED PROFESSIONAL ENGINEER

TRUONG VIET PHAM

No. 81809

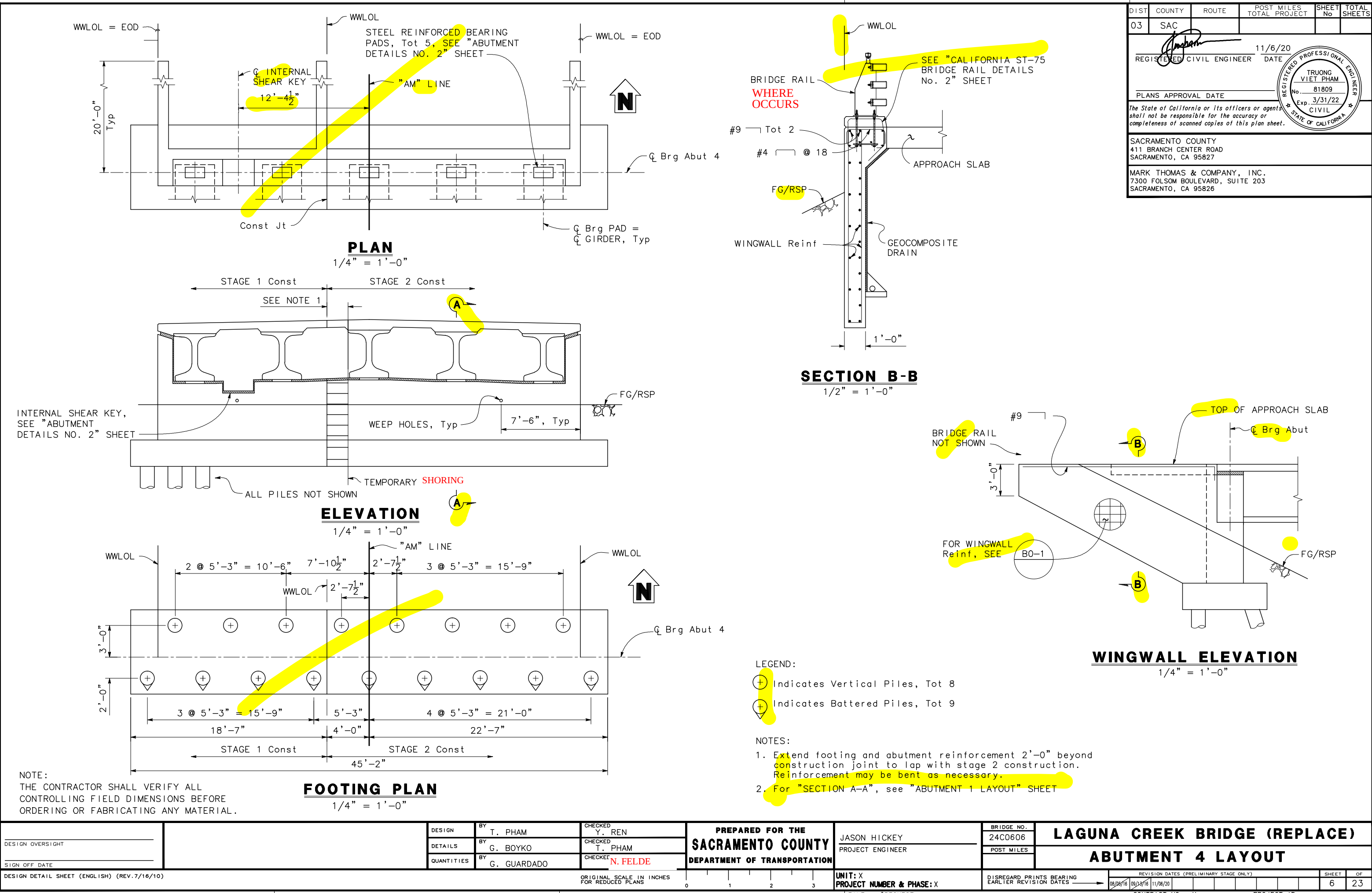
Exp. 3/31/22

CIVIL

STATE OF CALIFORNIA

DESIGN OVERSIGHT	SIGN OFF DATE	DESIGN	BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO.	24C0606	LAGUNA CREEK BRIDGE (REPLACE) ABUTMENT 1 LAYOUT	SHEET 5 OF 23	
		DETAILS	BY G. BOYKO	CHECKED T. PHAM			POST MILES				
		QUANTITIES	BY G. GUARDADO	CHECKED N. FELDE	UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)	CONTRACT NO.: X	PROJECT ID:
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)		ORIGINAL SCALE IN INCHES FOR REDUCED PLANS			FILE => \$REQUEST		11/28/13 04/24/14 04/28/14 04/28/14 12/02/14 08/28/15 08/18/16 09/18/16 11/06/20				

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11/6/20
DATE

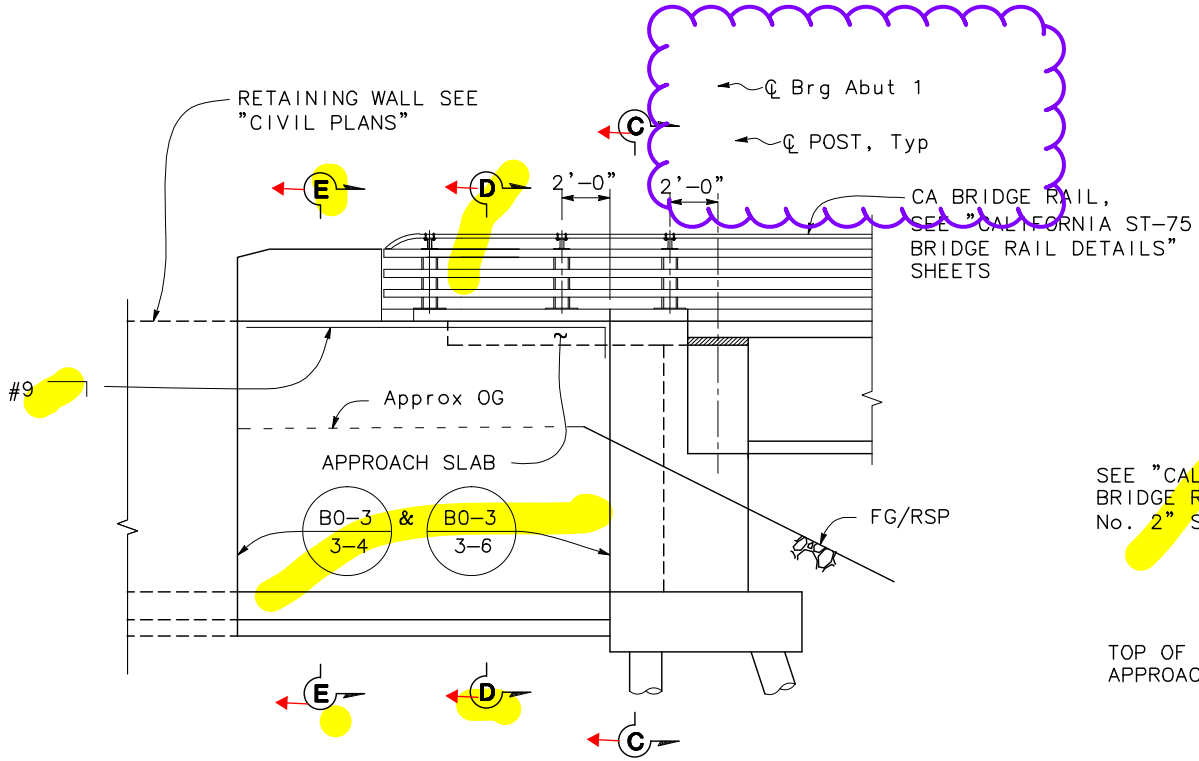
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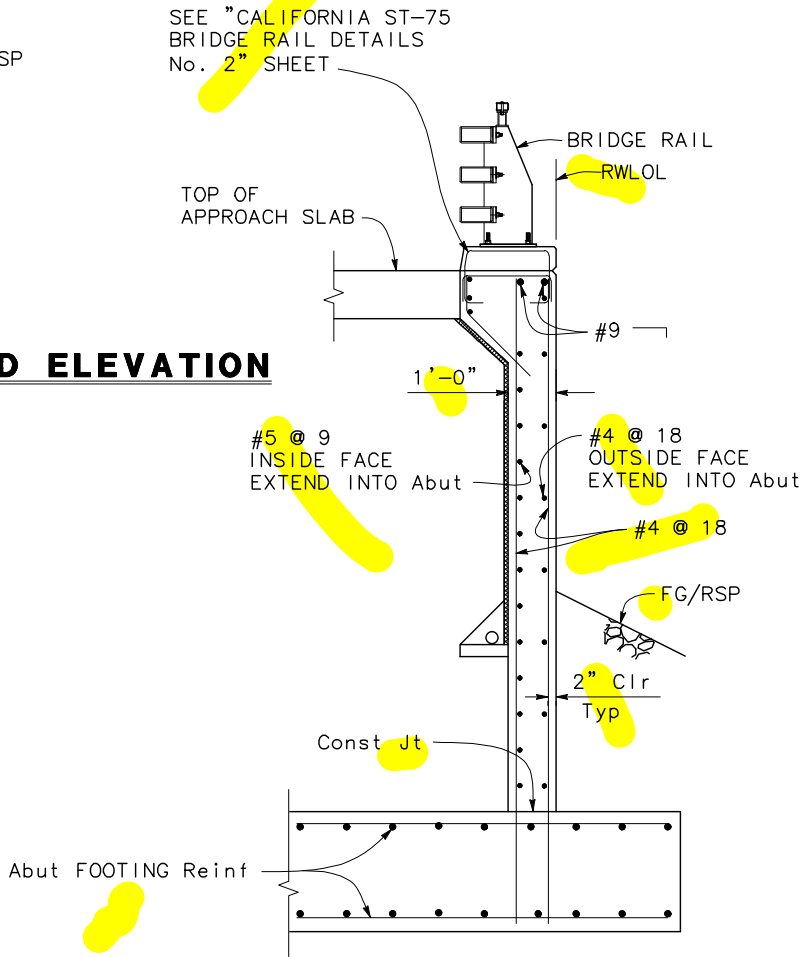
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SACRAMENTO, CA 95827

MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826

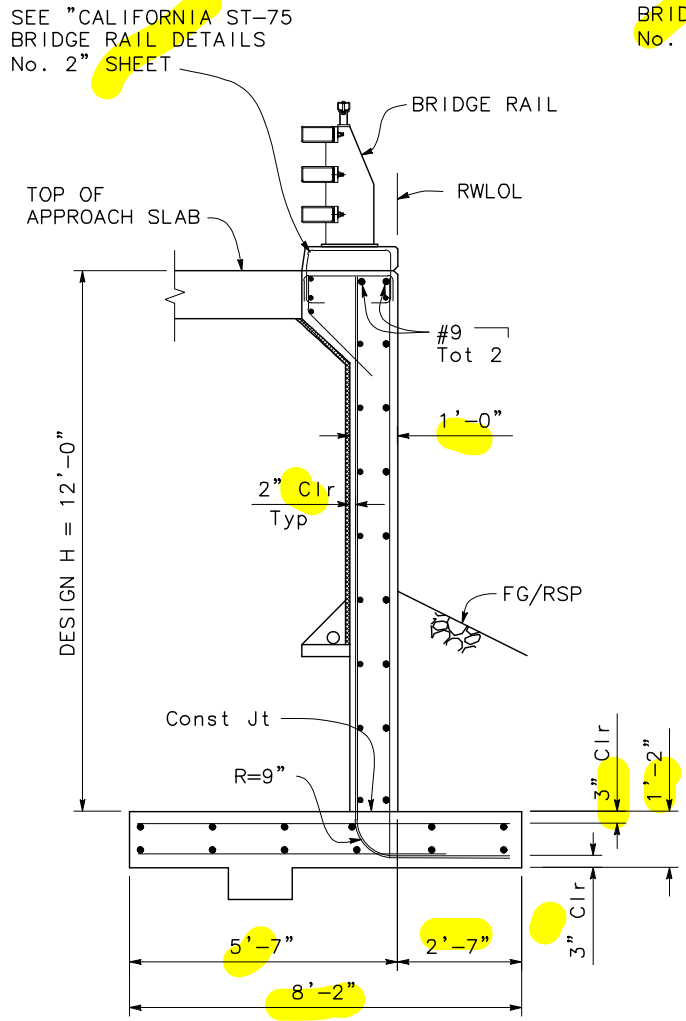


Abut 1 WEST RETAINING WALL - MIRRORED ELEVATION
1/4" = 1'-0"



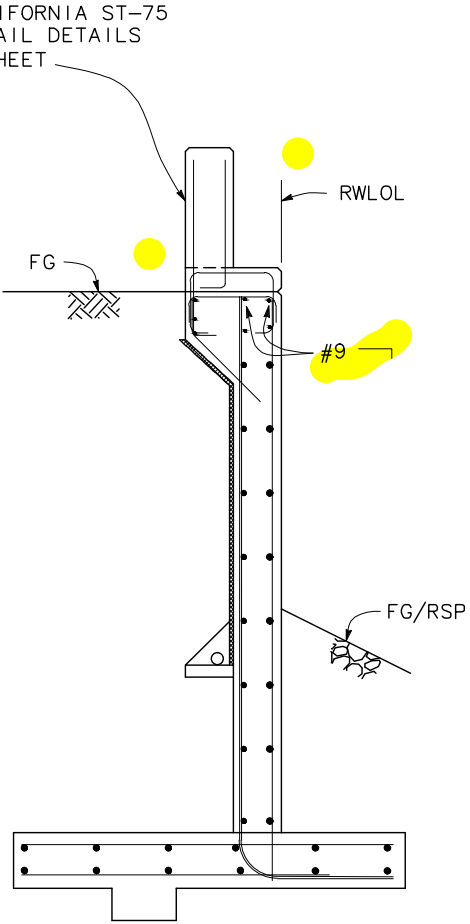
SECTION C-C
1/2" = 1'-0"

~~For details and reinforcement not shown, see RSP B11-77~~



SECTION D-D
1/2" = 1'-0"

~~For details and reinforcement not shown, see RSP B3-3A and RSP B11-77~~



SECTION E-E
1/2" = 1'-0"

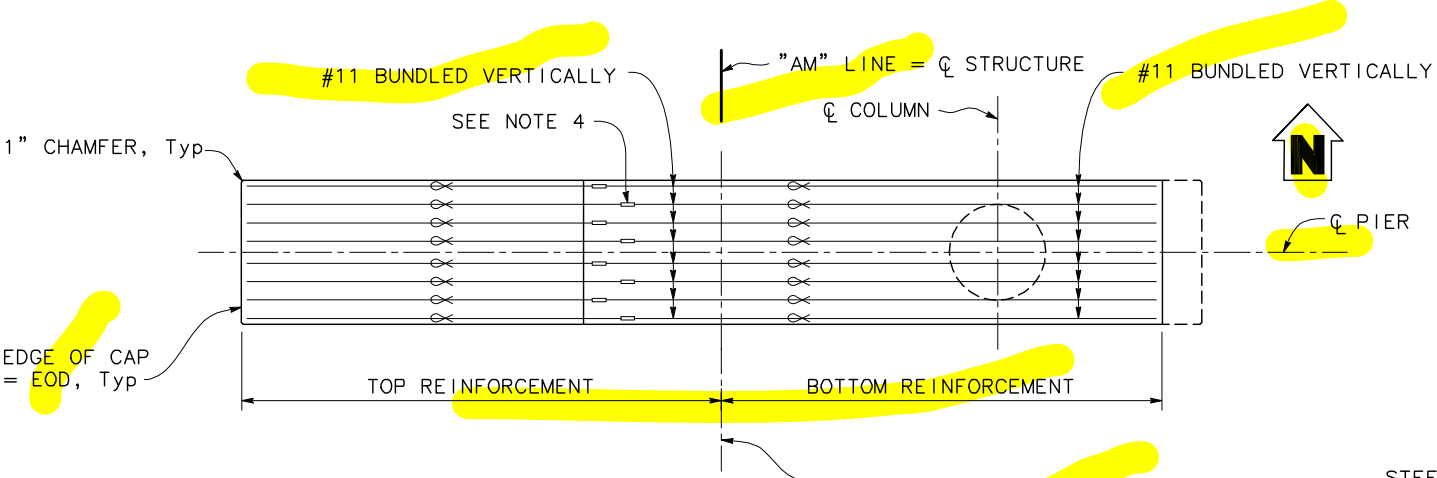
See "SECTION D-D" for information not shown

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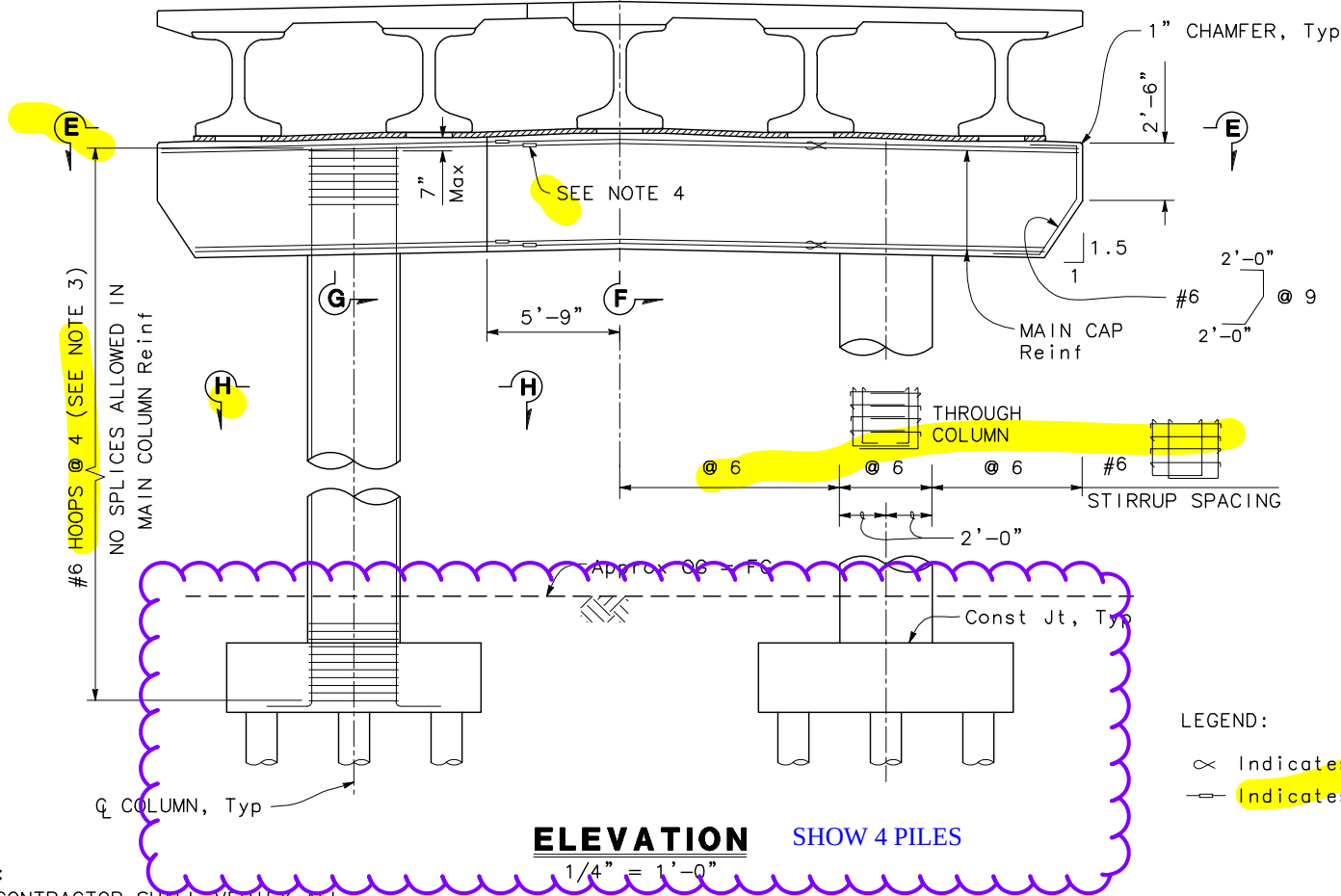
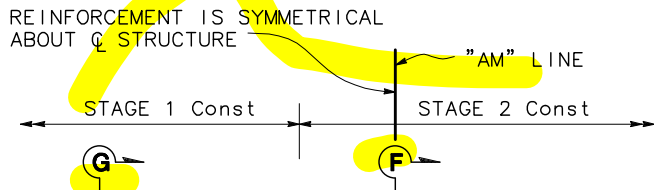
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DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)						ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	0 1 2 3	UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)						SHEET	OF
								FILE => \$REQUEST				CONTRACT NO.: X						7	23
												PROJECT ID:							

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			11/6/20		
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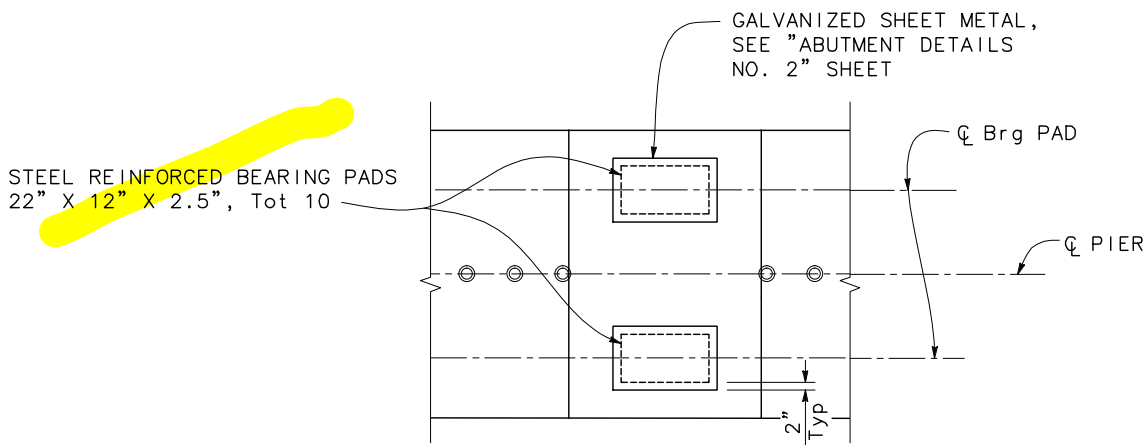
PLAN
1/4" = 1'-0"



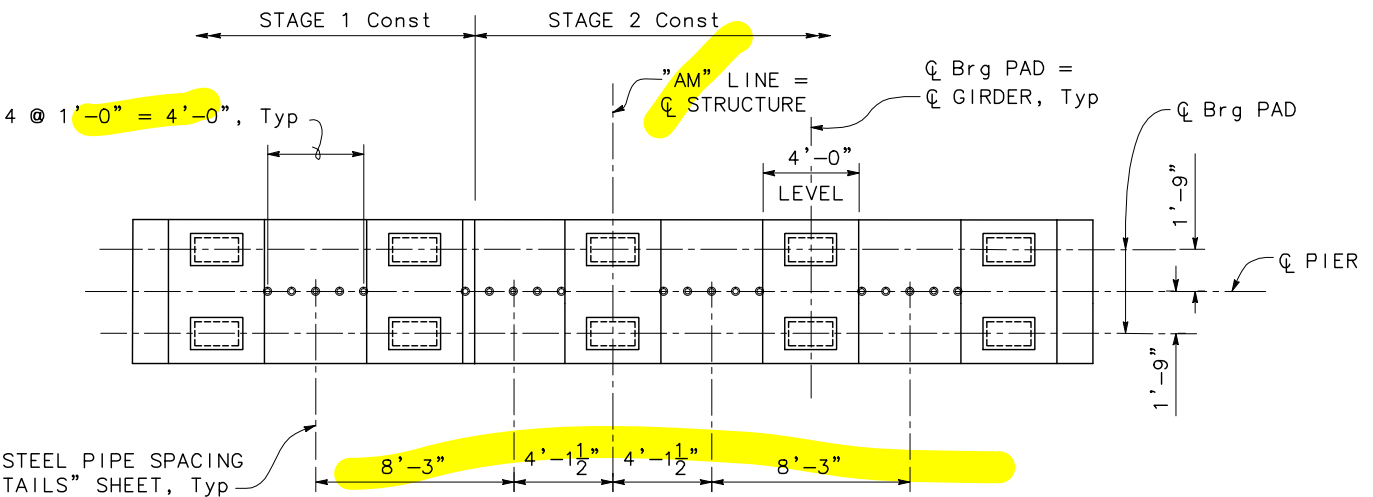
ELEVATION SHOW 4 PILES
1/4" = 1'-0"

- LEGEND:
- ∞ Indicates bundled bars
 - Indicates mechanical splice

NOTE:
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PIER BEARING PAD DETAILS
1/2" = 1'-0"



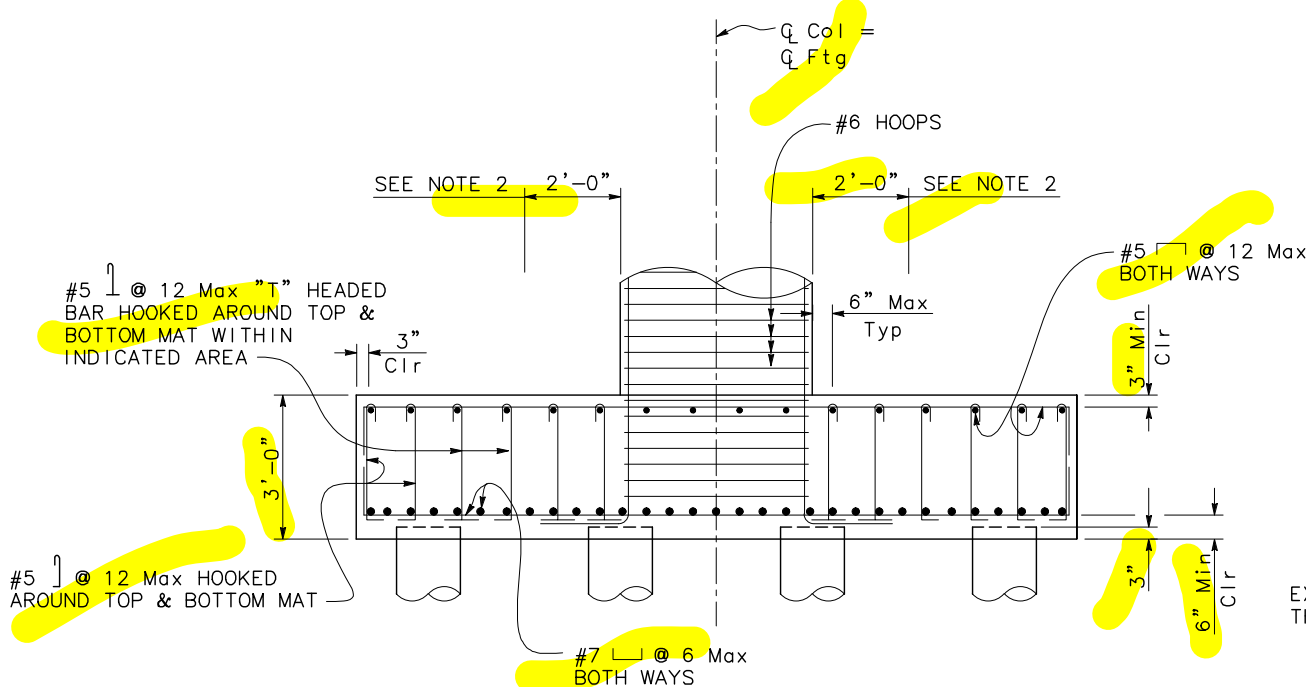
SECTION E-E
1/4" = 1'-0"

- NOTES:
- For "SECTION F-F", "SECTION G-G" and "SECTION H-H", see "PIER DETAILS" sheet.
 - Pier 2 shown, Pier 3 similar.
 - Hoops shall be ultimate spliced.
 - Mechanical splice, stagger top, bottom and abjacent bars.

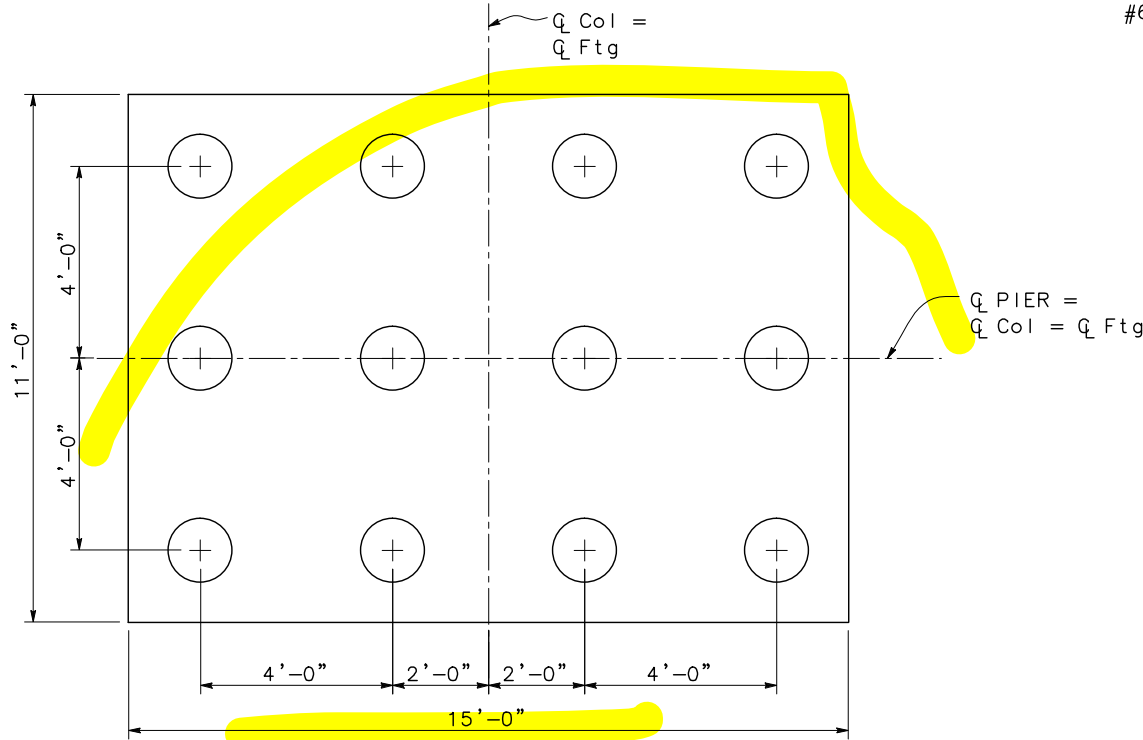
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				FILE => \$REQUEST	CONTRACT NO.: X		PROJECT ID:

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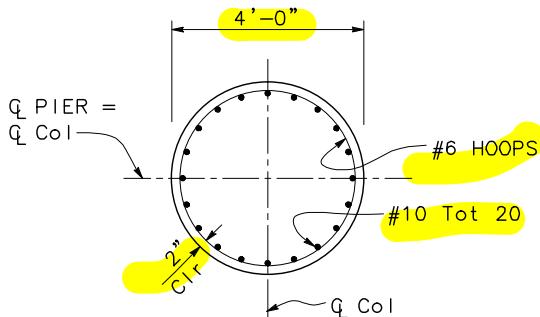


FOOTING ELEVATION
1/2" = 1'-0"

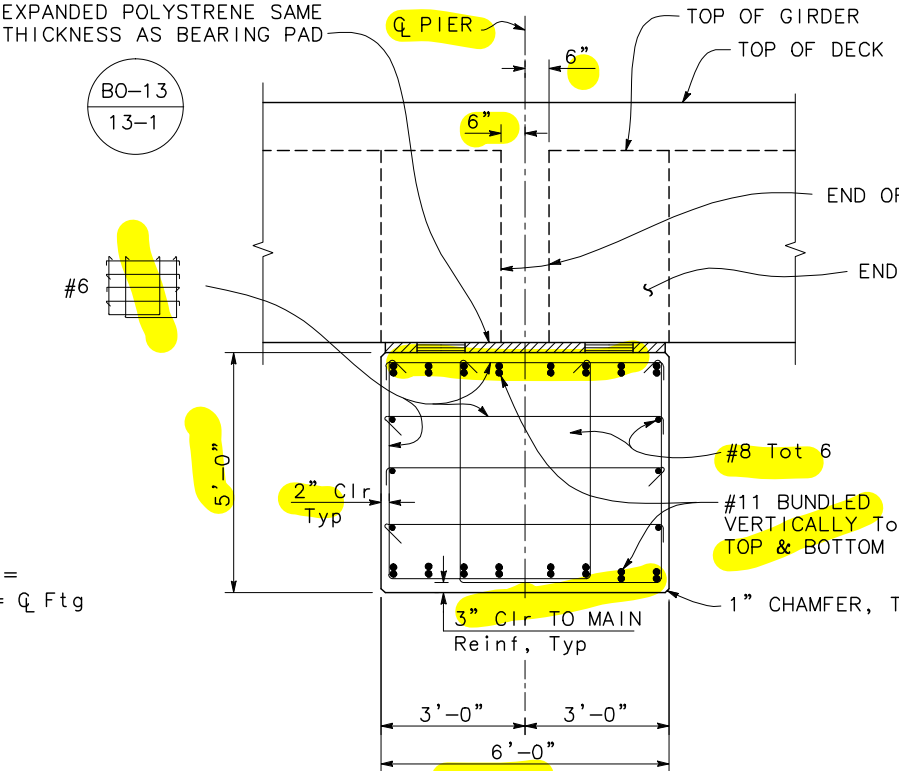


FOOTING PLAN
1/2" = 1'-0"

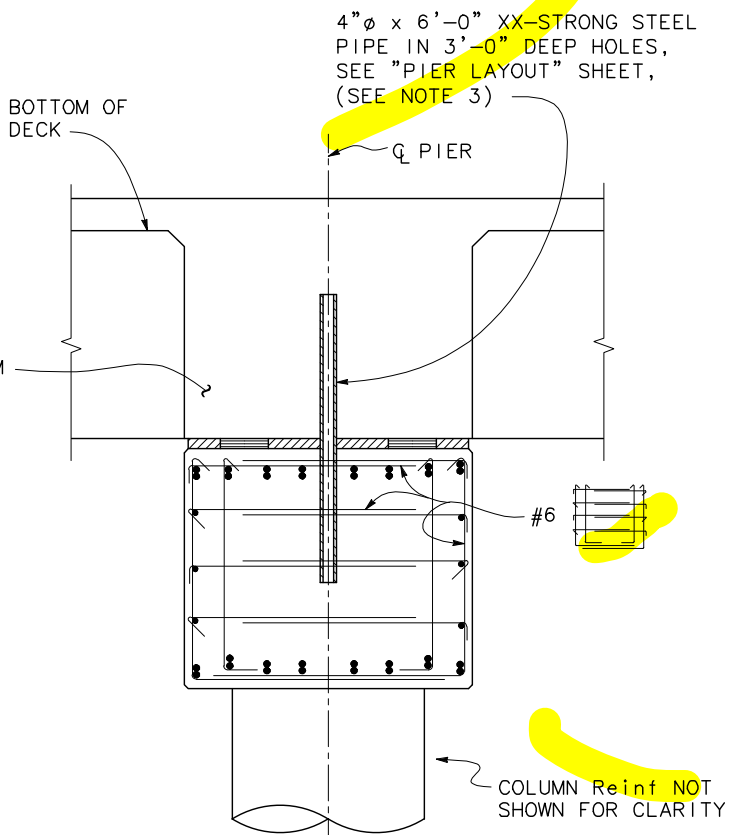
NOTE:
THE CONTRACTOR SHALL VERIFY ALL
CONTROLLING FIELD DIMENSIONS BEFORE
ORDERING OR FABRICATING ANY MATERIAL.



SECTION H-H
1/2" = 1'-0"



SECTION F-F
1/2" = 1'-0"



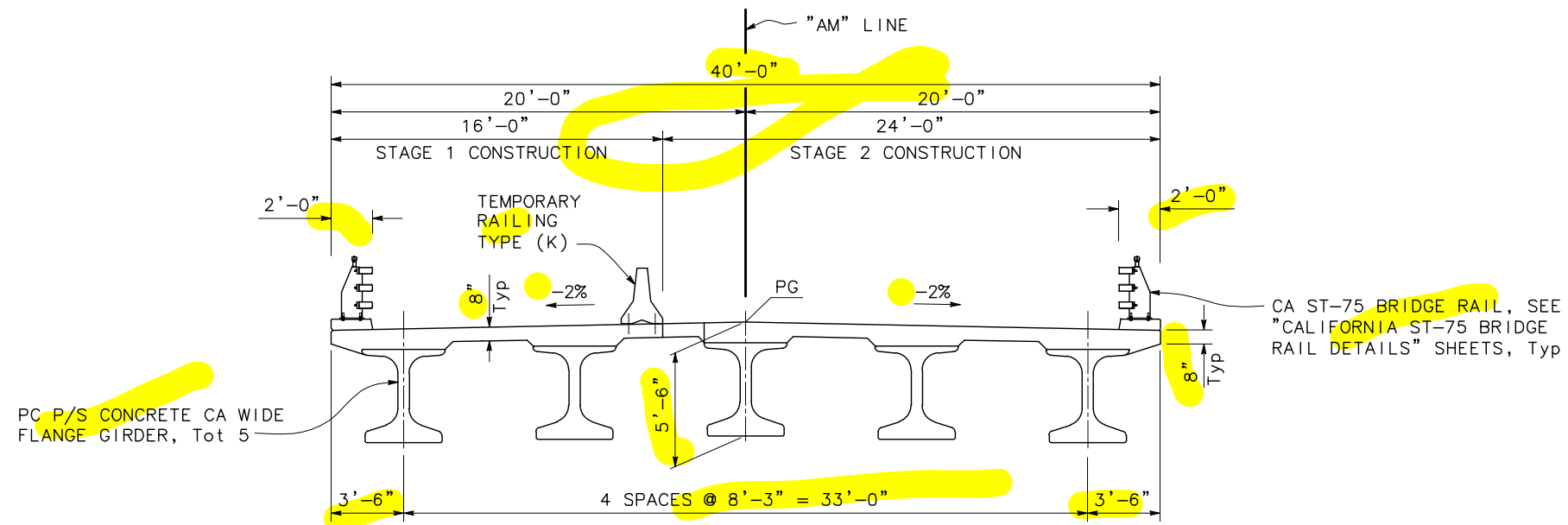
SECTION G-G
1/2" = 1'-0"

NOTE:
See "SECTION F-F" for information not shown.

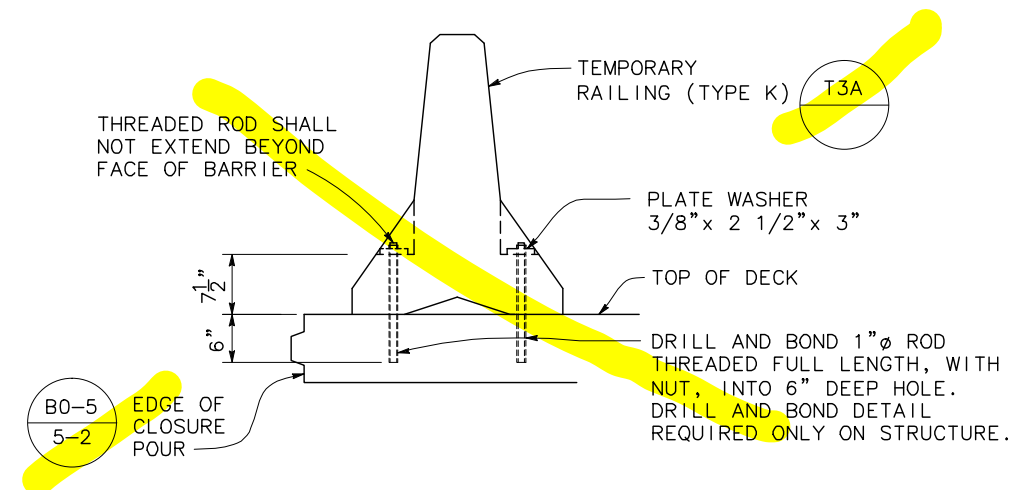
- NOTES:
- For locations of "SECTION F-F", "SECTION G-G" and "SECTION H-H", see "PIER LAYOUT" sheet.
 - Location of "T" headed reinforcement around column.
 - Fill pipe with grout.

DESIGN OVERSIGHT	SIGN OFF DATE	DESIGN	BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO.	24C0606	LAGUNA CREEK BRIDGE (REPLACE)
		DETAILS	BY G. BOYKO	CHECKED T. PHAM			POST MILES		
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)		QUANTITIES	BY G. GUARDADO	CHECKED N. FELDE	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)		SHEET 10 OF 23
ORIGINAL SCALE IN INCHES FOR REDUCED PLANS					0 1 2 3	FILE => \$REQUEST	CONTRACT NO.: X		PROJECT ID:

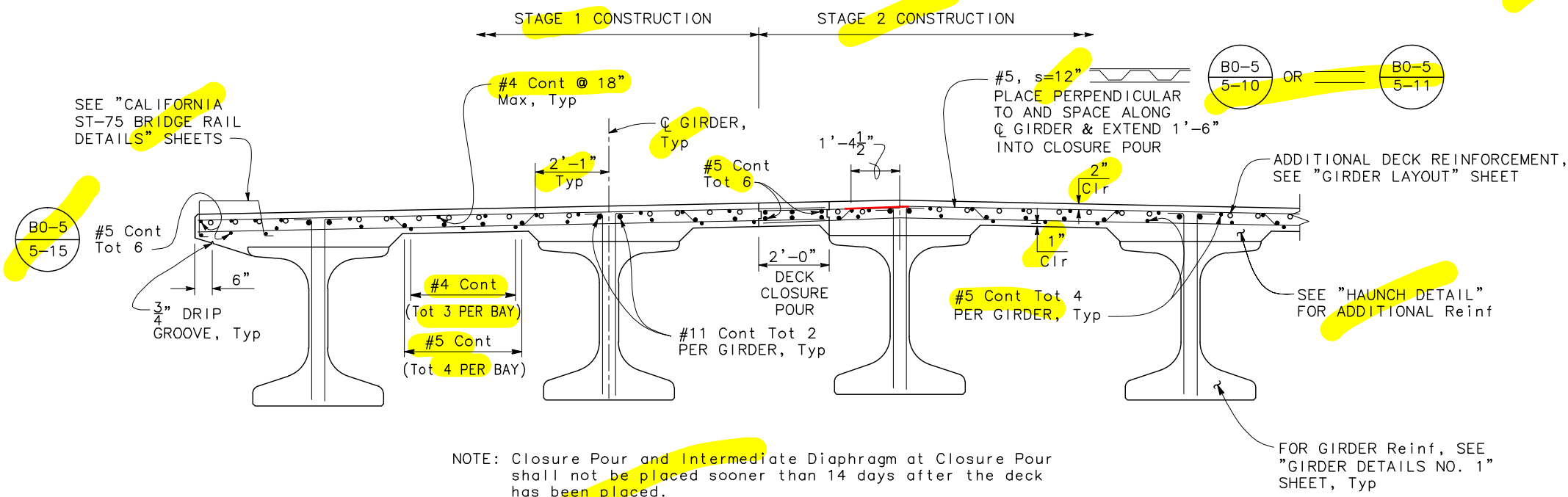
DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
			11/6/20		
REGISTERED CIVIL ENGINEER			DATE		
PLANS APPROVAL DATE					
<i>The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.</i>					
SACRAMENTO COUNTY 411 BURNER CENTER ROAD SACRAMENTO, CA 95827					
MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					



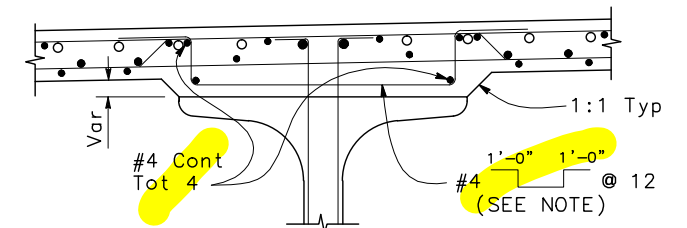
TYPICAL SECTION
1/4" = 1'-0"



TEMPORARY RAILING (TYPE K)
ATTACHMENT DETAILS
NO SCALE



PART TYPICAL SECTION
1/2" = 1'-0"



HAUNCH DETAIL

NO SCALE

NOTE: Reinf shown is in addition to "PART TYPICAL SECTION" Reinf

NOTE:
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DESIGN OVERSIGHT				DESIGN	BY	T. PHAM	CHECKED	Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION		JASON HICKEY PROJECT ENGINEER		BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)																									
				DETAILS	BY	G. BOYKO	CHECKED	T. PHAM					POST MILES	TYPICAL SECTION																									
SIGN OFF DATE				QUANTITIES	BY	G. GUARDADO	CHECKED	N. FELDE																															
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)								ORIGINAL SCALE IN INCHES FOR REDUCED PLANS		0 1 2 3		UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY) 11/04/13 04/28/14 04/28/14 05/28/15 07/28/16 08/17/16 09/13/17 11/06/20								SHEET OF 11 23															
										FILE ==> \$REQUEST										CONTRACT NO.: X										PROJECT ID:									

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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			11/6/20	DATE	
PLANS APPROVAL DATE			11/6/20		
The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.			REGISTERED PROFESSIONAL ENGINEER		
SACRAMENTO COUNTY			No. 81809		
411 BRANCH CENTER ROAD			Exp. 3/31/22		
SACRAMENTO, CA 95827			CIVIL		
MARK THOMAS & COMPANY, INC.					
7300 FOLSOM BOULEVARD, SUITE 203					
SACRAMENTO, CA 95826					



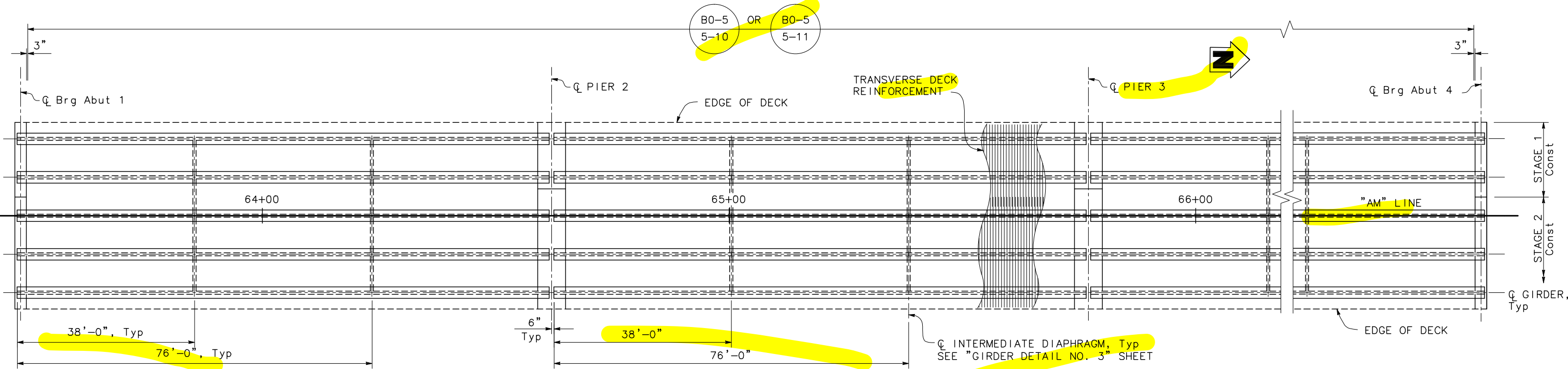
ADDITIONAL DECK REINFORCEMENT

3/16" = 1'-0"

#11 x 40'-0" Tot 8
BETWEEN GIRDERS, Typ
(SEE NOTE 2)

NOTES:

1. Place Additional Deck Reinforcement between longitudinal deck reinforcement as shown on "TYPICAL SECTION" sheet.
2. No slices allowed for #11 bars.
3. No splices allowed 25'-0" each side of Q Pier for #11 Cont bars.
4. Additional Deck Reinforcement at Pier 2 shown, Pier 3 similar.



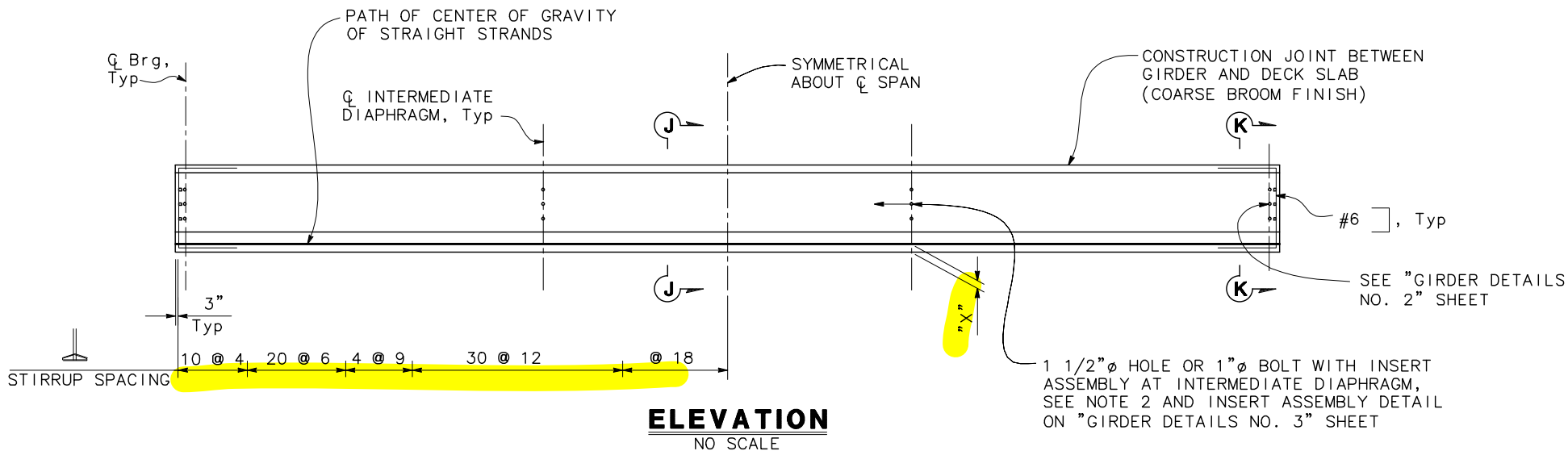
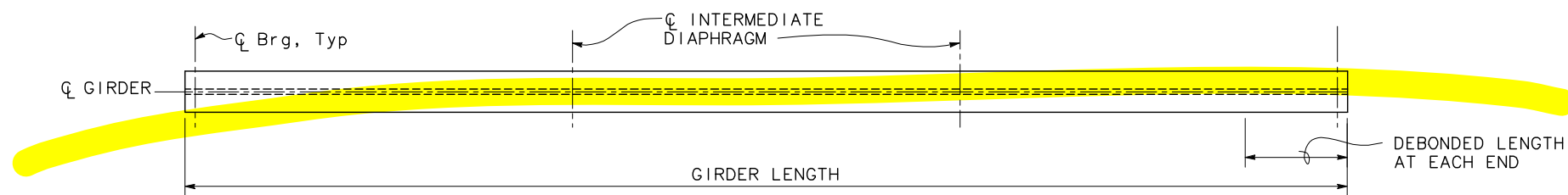
GIRDER LAYOUT

3/32" = 1'-0"

NOTE:
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ORDERING OR FABRICATING ANY MATERIAL.

DESIGN OVERSIGHT		DESIGN		BY	T. PHAM	CHECKED	Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION		BRIDGE NO.		LAGUNA CREEK BRIDGE (REPLACE)															
SIGN OFF DATE		DETAILS		BY	G. BOYKO	CHECKED	T. PHAM			JASON HICKEY PROJECT ENGINEER		24C0606		GIRDER LAYOUT													
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)		QUANTITIES		BY	G. GUARDADO	CHECKED	N. FELDE			POST MILES																	
						ORIGINAL SCALE IN INCHES FOR REDUCED PLANS		0 1 2 3		UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)										SHEET		OF	
														11/28/13 03/28/14 04/28/14 05/07/14 12/02/14 05/18/15 07/28/16 08/11/16 11/06/20										12		23	

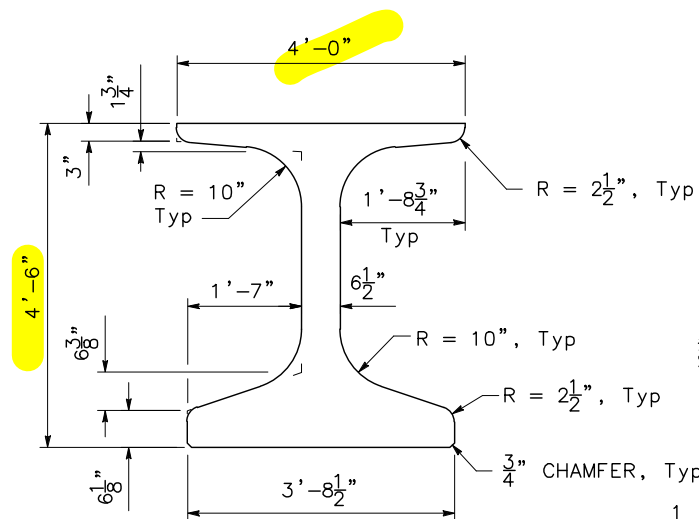
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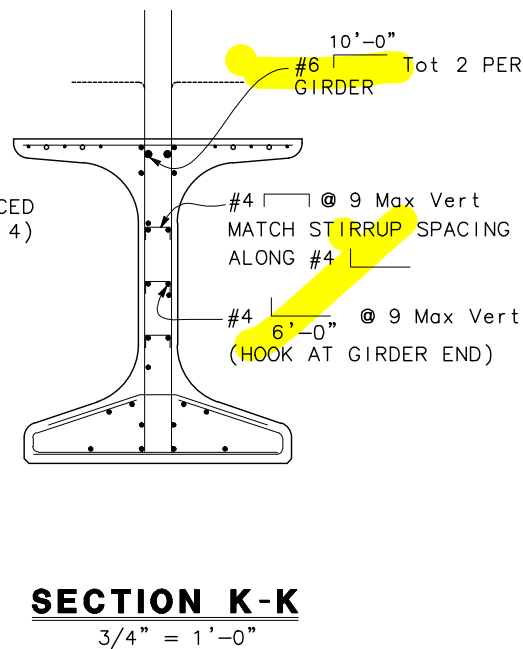
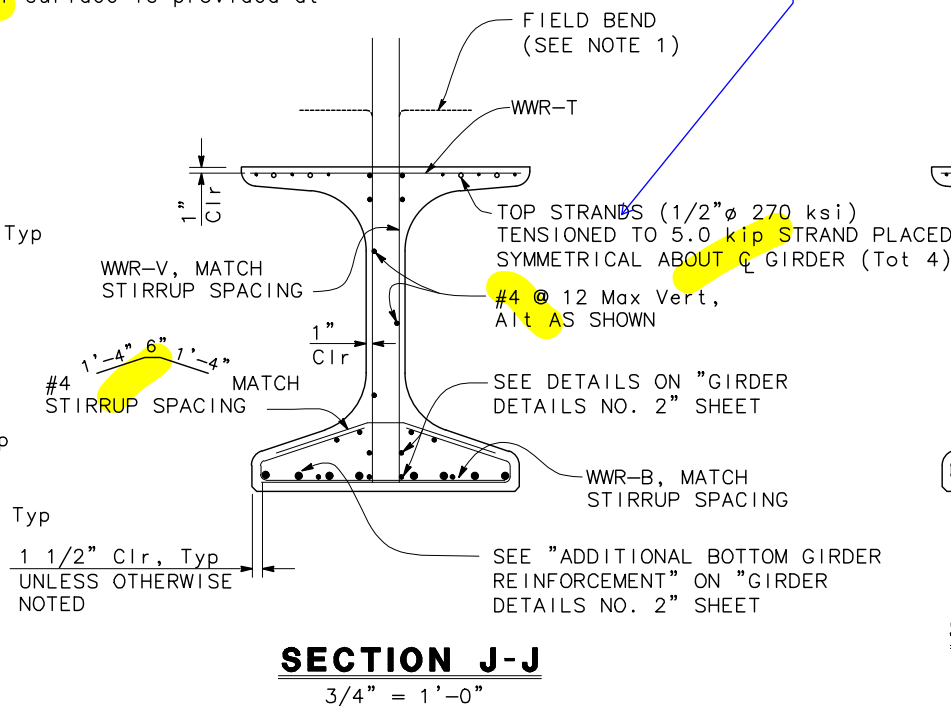
NOTES:

- Girder stirrups shall be field bent after girder erection.
- Insert assemblies shall be parallel to ϕ Intermediate Diaphragms.
- Form holes shall be parallel to ϕ Brg.
- Girder ends to be cast such that a level surface is provided at bearing pads.

why PT needed? OK to keep



NOTE:
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TOTAL NO. OF 0.6" STRANDS	NO. OF DEBONDED STRANDS	DEBONDED LENGTH
36	12	9', 12', 15'

* ALTERNATE DEBONDED LENGTH AND MIRRORED ABOUT ϕ GIRDER

PER GIRDER:

GIRDER LENGTH	GIRDER DEPTH	"X" (in)	JACKING FORCE (P) (kips)	CONCRETE STRENGTH (ksi)		MIDSPAN DEAD LOAD DEFLECTION (in)	
				f'ci	f'c	DECK	RAIL
114'-0"	4'-6"	4.05	1582	7.0	10.0	1.70	0.15

PRESTRESSING NOTES

0.04 ok

- The Jacking Force (P) is the jacking force required at the point of control along the span. The jacking force does not include any fabrication specific losses
- The maximum tensile stress in the prestressing steel upon release shall not exceed 75% of the specified minimum ultimate tensile strength of the prestressing steel
- The maximum temporary tensile stress (jacking stress) in the prestressing steel shall not exceed 80% of the specified minimum ultimate tensile strength of the prestressing steel
- Concrete strength:
f'ci is at time of initial stressing
f'c is at 28 days
- Deflection components are informational and will be used to set screed line elevations
- Screed line elevations for deck concrete will be determined by the Engineer
- Prestressing strand shall be 270 ksi low relaxation
- No more than 33% of the total number of strands and 50% of the strands per horizontal row may be debonded
- Strand locations may be adjusted as approved by the Engineer

DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE) GIRDER DETAILS NO. 1	
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE	POST MILES		
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)			UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)
			ORIGINAL SCALE IN INCHES FOR REDUCED PLANS		CONTRACT NO.: X		PROJECT ID:
			FILE => \$REQUEST		SHEET 13 OF 23		

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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				


REGISTERED CIVIL ENGINEER

11/6/20
DATE

REGISTERED PROFESSIONAL ENGINEER

TRUONG VIET PHAM

No. 81809

Exp. 3/31/22

CIVIL

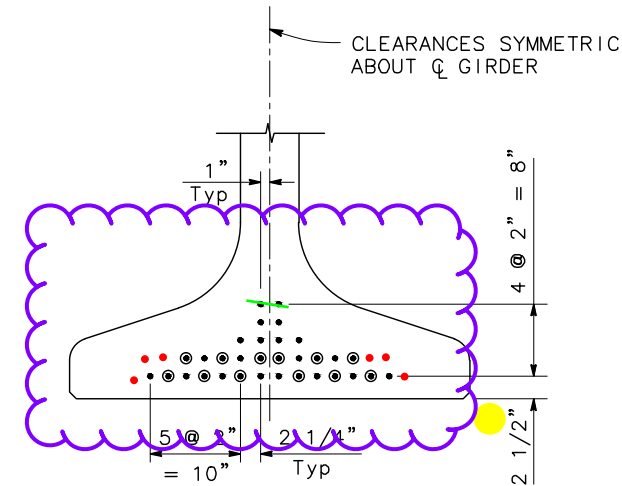
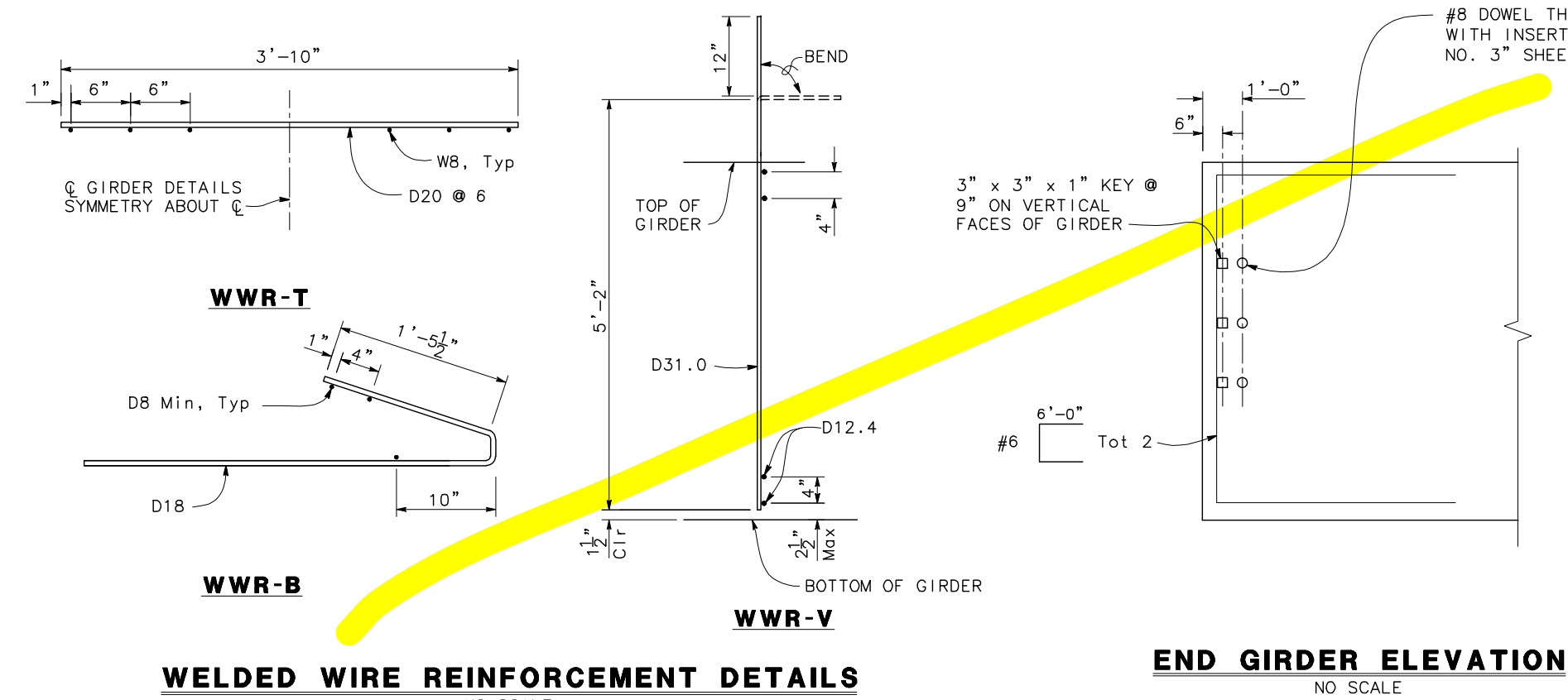
STATE OF CALIFORNIA

PLANS APPROVAL DATE

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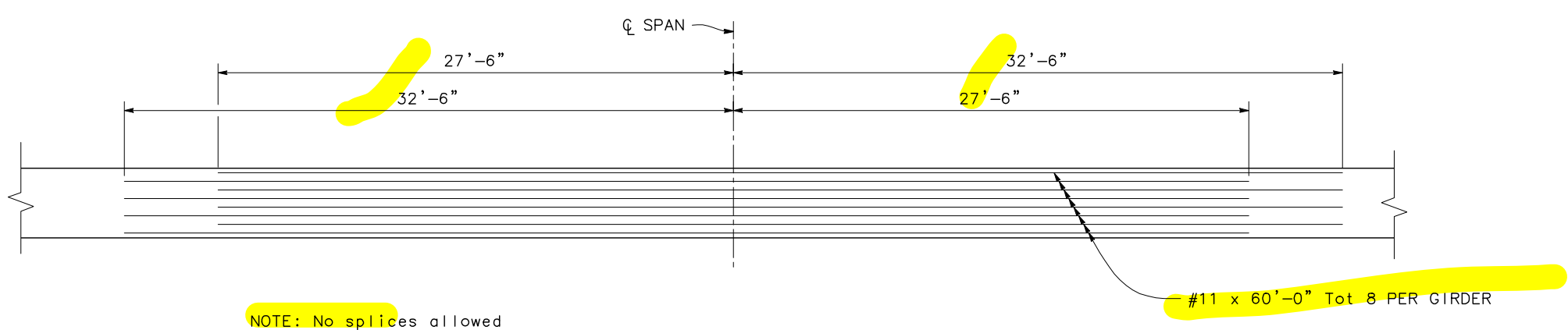
SACRAMENTO COUNTY
411 BRANCH CENTER ROAD
SACRAMENTO, CA 95827

MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826



STRANDS LAYOUT
NO SCALE

- LEGEND:
- Denotes continuously bonded strand location
 - ⊙ Denotes permissible debonded strand location
- NOTES:
- Clearances shown are for 0.6" nominal diameter strands.
 - Approval by the Engineer is required for deviation.



NOTE: No splices allowed

ADDITIONAL BOTTOM GIRDER REINFORCEMENT
NO SCALE

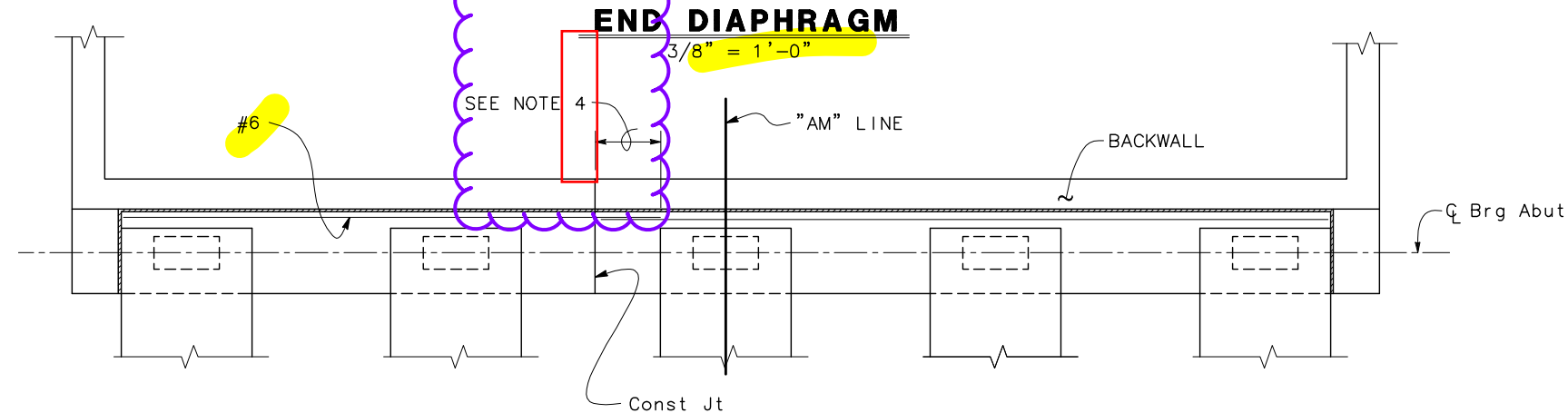
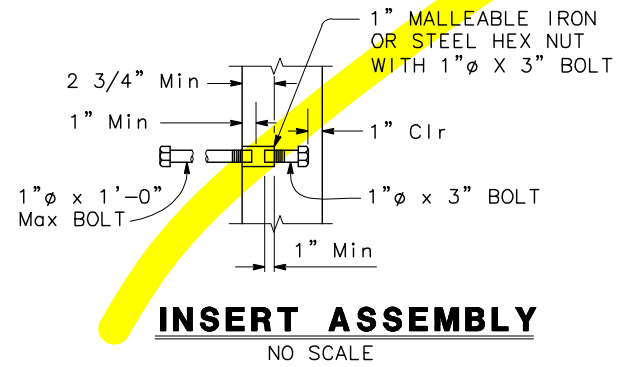
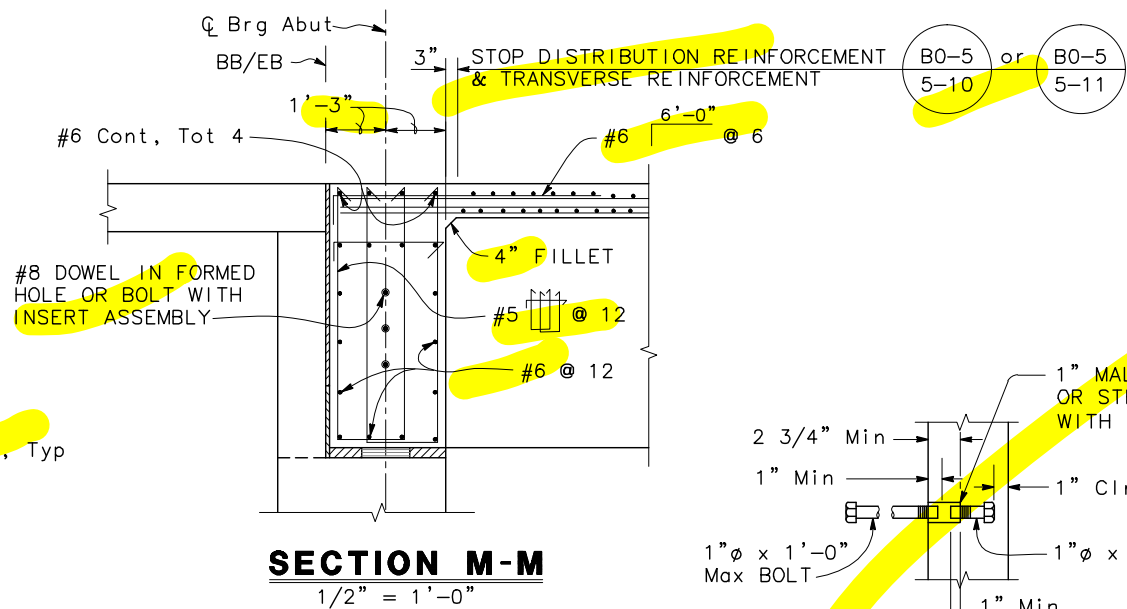
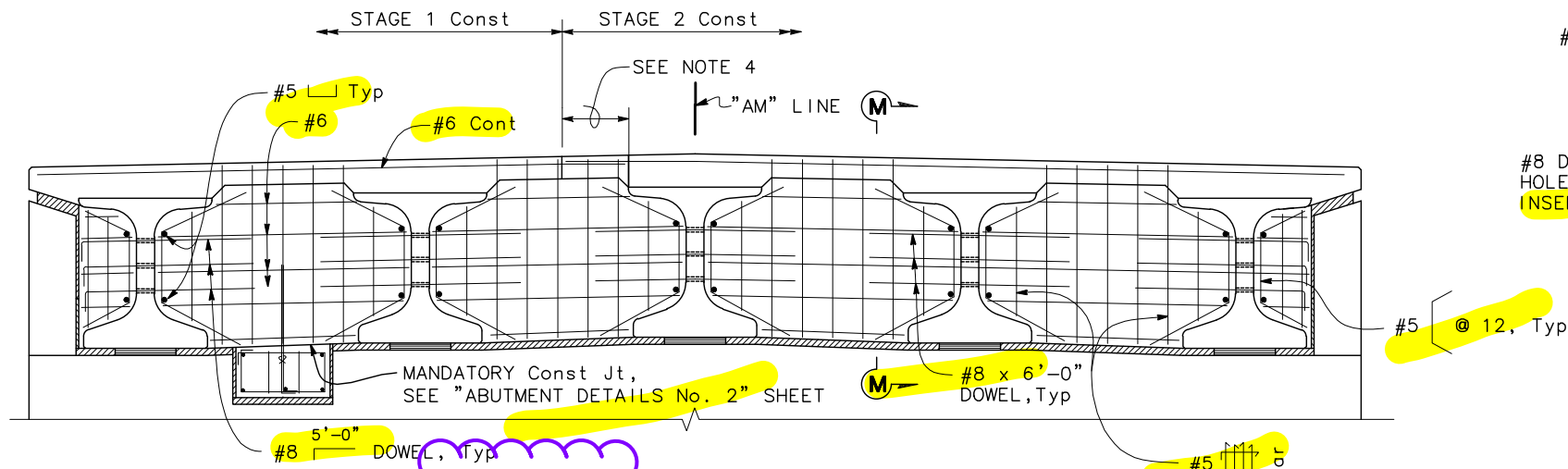
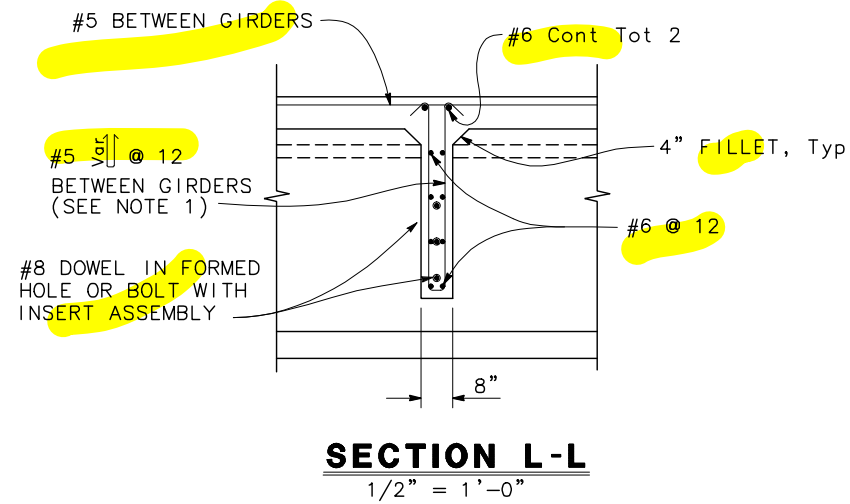
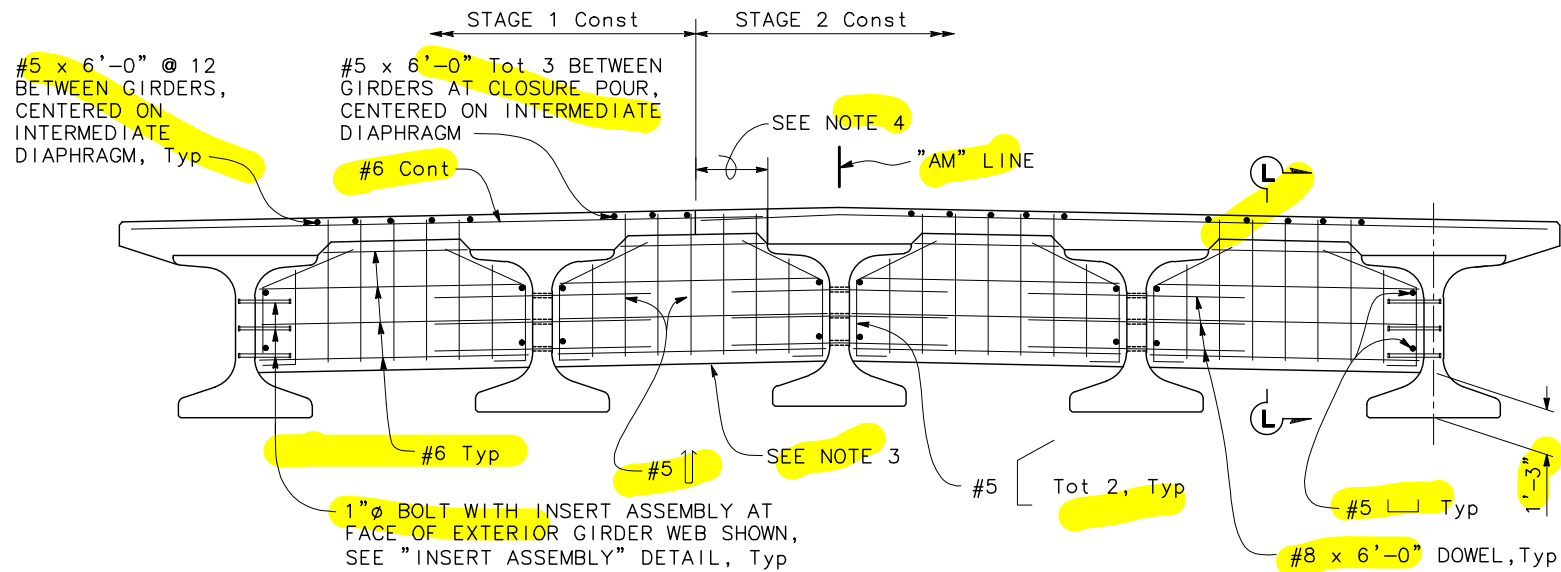
NOTE:
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DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE) GIRDER DETAILS NO. 2				
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM		POST MILES						
	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE								
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)			ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)		SHEET 14	OF 23
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TIME PLOTTED => \$TIME
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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			DATE	11/6/20	
PLANS APPROVAL DATE					
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SACRAMENTO COUNTY 411 BRANCH CENTER ROAD SACRAMENTO, CA 95827			REGISTERED PROFESSIONAL ENGINEER No. 81809 Exp. 3/31/22 CIVIL STATE OF CALIFORNIA		
MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					



NOTE:
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Note: Abut 4 shown, Abut 1 similar.

NOTES:

- Hook reinforcement inward below girder top flange.
- Intermediate diaphragm may be vertical or normal to deck grade.
- Intermediate diaphragm at closure pour shall be constructed with closure pour.
- Extend reinforcement 2'-0" beyond construction joint to lap with stage 2 construction. Reinforcement may be bent as necessary.

DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)	
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM			POST MILES	GIRDER DETAILS NO. 3	
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)	SHEET 15	OF 23
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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				


REGISTERED CIVIL ENGINEER

11/6/20
DATE

PLANS APPROVAL DATE

TRUONG VIET PHAM
No. 81809
Exp. 3/31/22
CIVIL
STATE OF CALIFORNIA

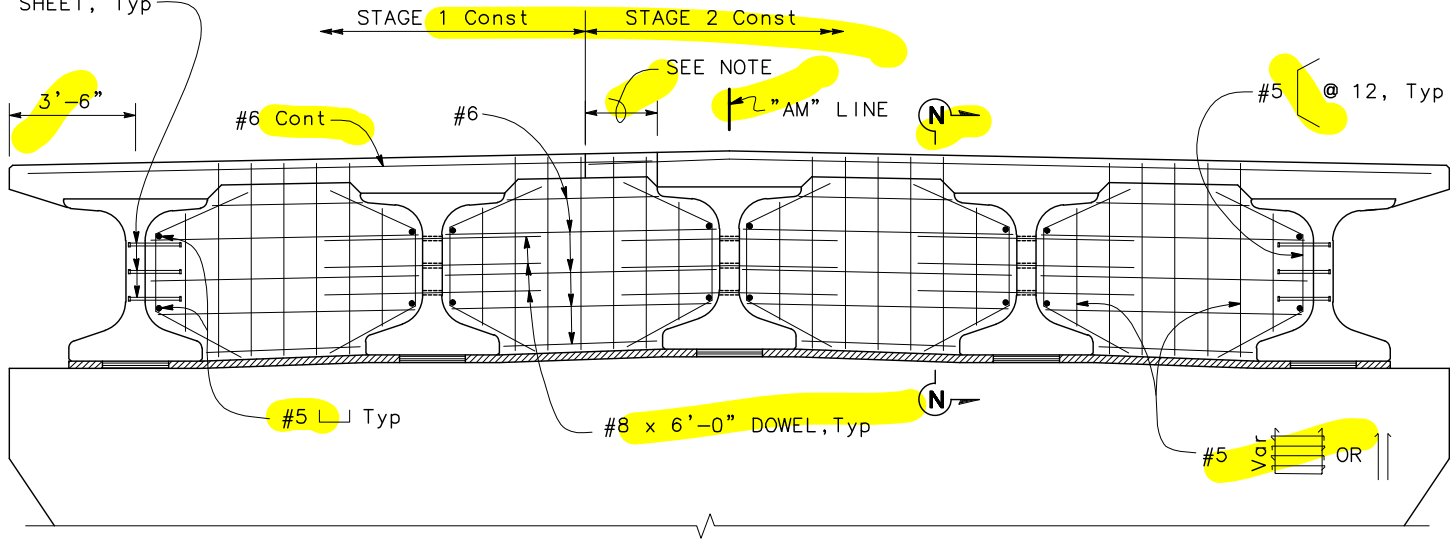
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SACRAMENTO, CA 95827

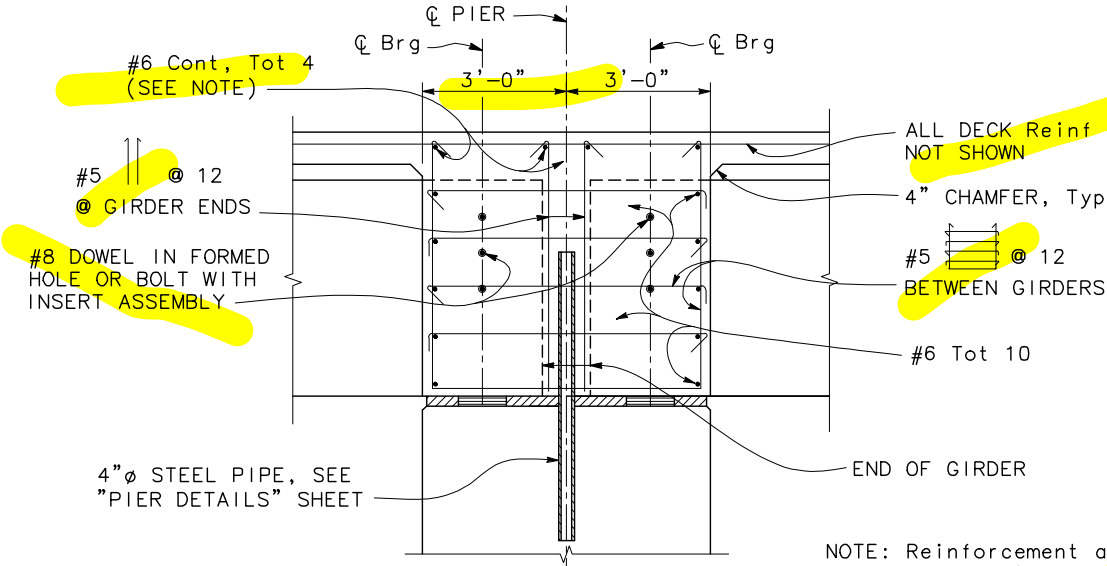
MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826

NOTE:
Extend reinforcement 2'-0" beyond construction joint to lap with stage 2 construction. Reinforcement may be bent as necessary.

1"Ø BOLT WITH INSERT ASSEMBLY AT FACE OF EXTERIOR GIRDER WEB SHOWN, SEE "INSERT ASSEMBLY" DETAIL ON "GIRDER DETAILS NO.3" SHEET, Typ



PIER ELEVATION
3/8" = 1'-0"



SECTION N-N
1/2" = 1'-0"

NOTE: Reinforcement at Pier 2 shown, Pier 3 similar.

NOTE:
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DESIGN OVERSIGHT SIGN OFF DATE	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE) GIRDER DETAILS NO. 4	
	DETAILS BY G. BOYKO	CHECKED T. PHAM			POST MILES		
	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE					
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)			ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY) 07/28/16 08/11/16 11/06/20	SHEET 16 OF 23

FILE => \$REQUEST

CONTRACT NO.: X

PROJECT ID:

TIME PLOTTED => \$TIME
DATE PLOTTED => \$DATE
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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			DATE	11/6/20	
PLANS APPROVAL DATE					
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SACRAMENTO COUNTY 411 BRANCH CENTER ROAD SACRAMENTO, CA 95827					
MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					

REGISTERED PROFESSIONAL ENGINEER

TRUONG VIET PHAM

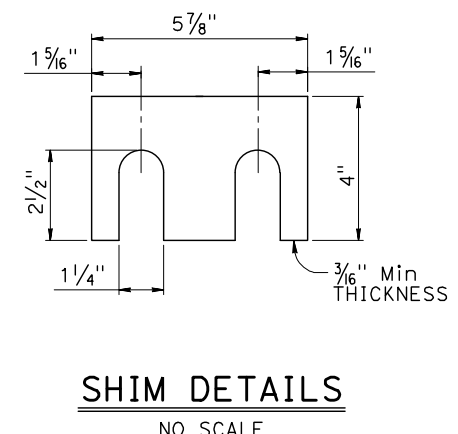
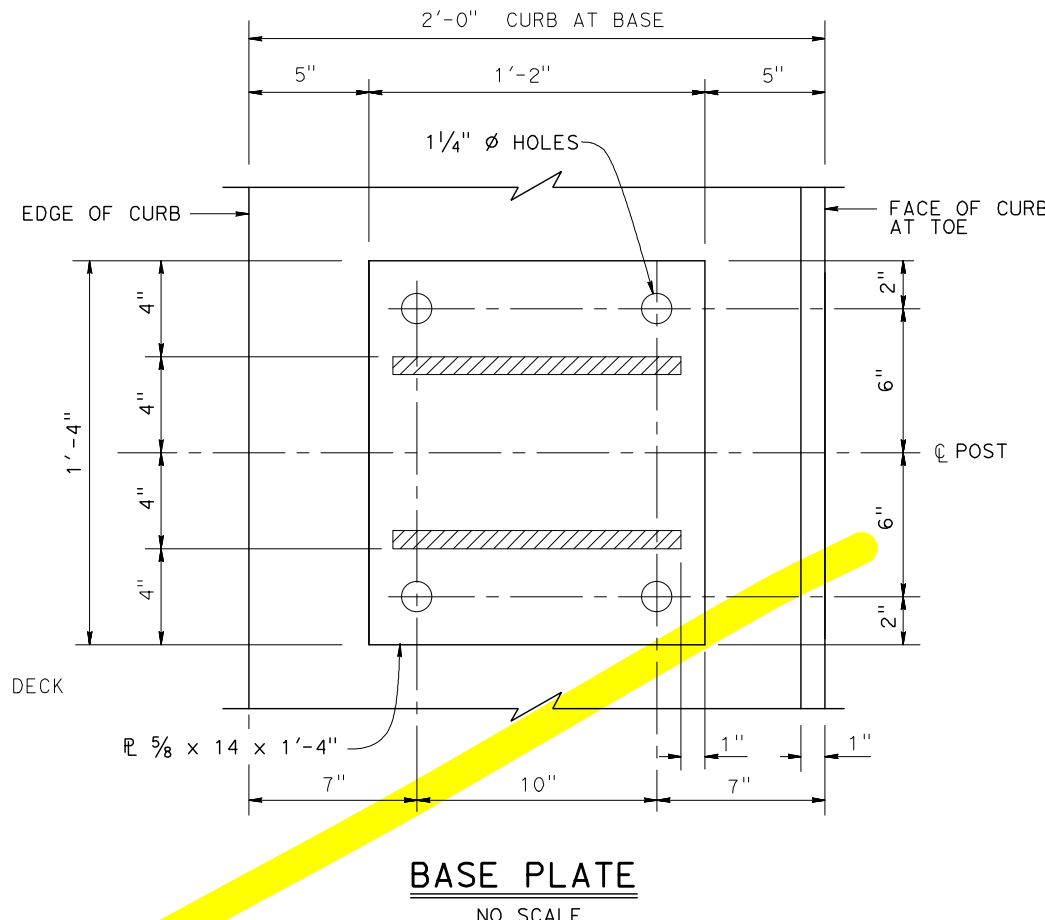
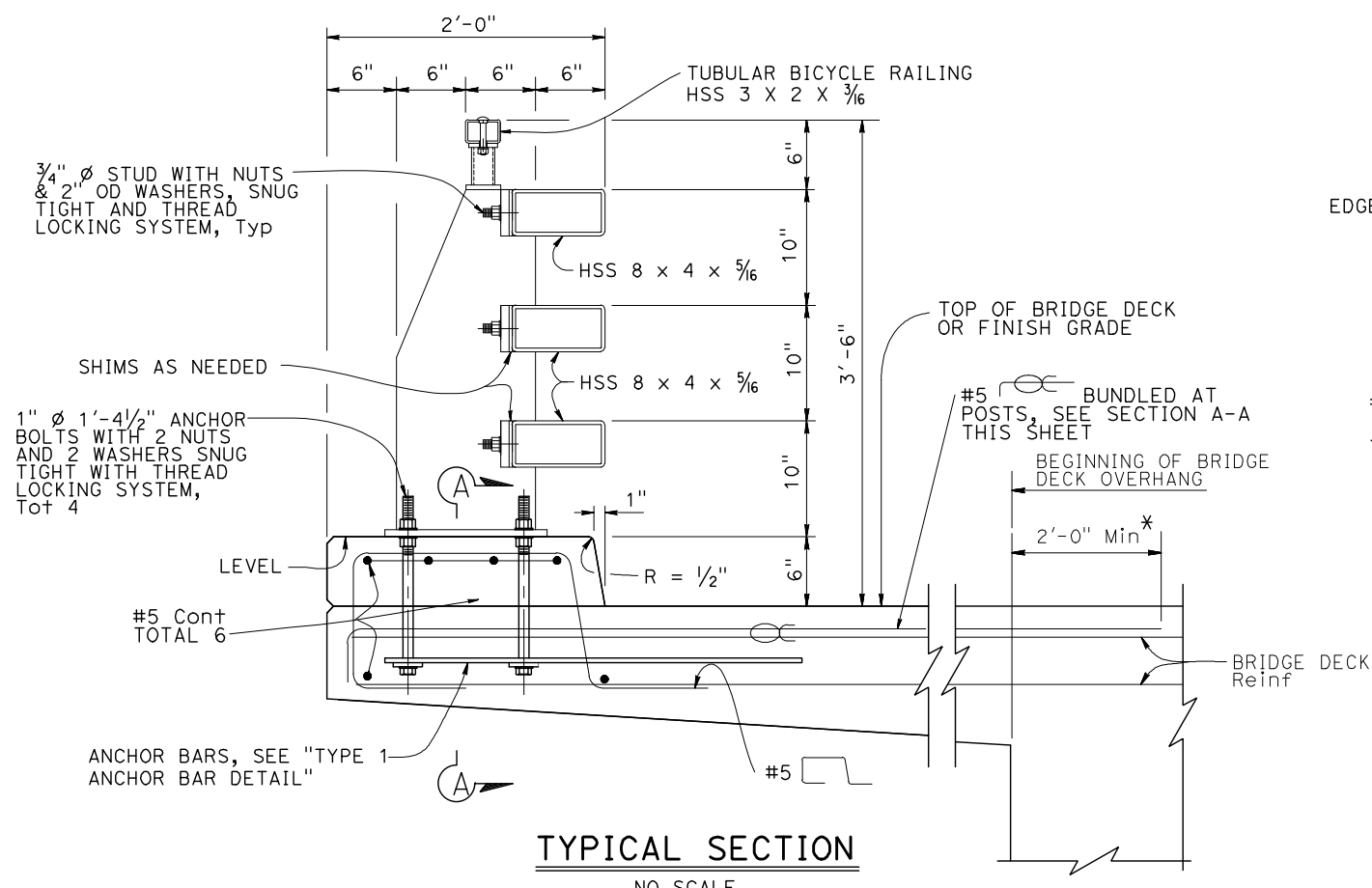
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Exp. 3/31/22

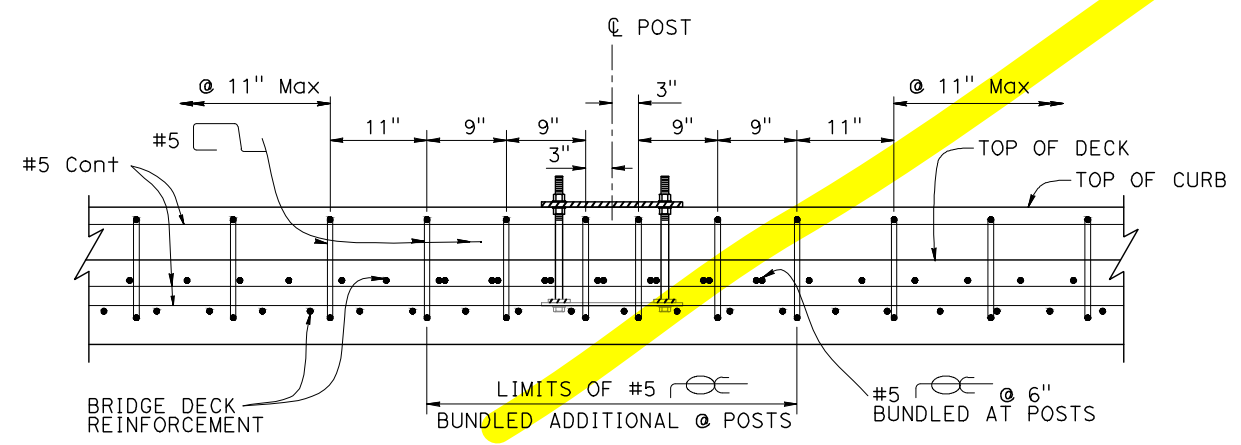
CIVIL

STATE OF CALIFORNIA

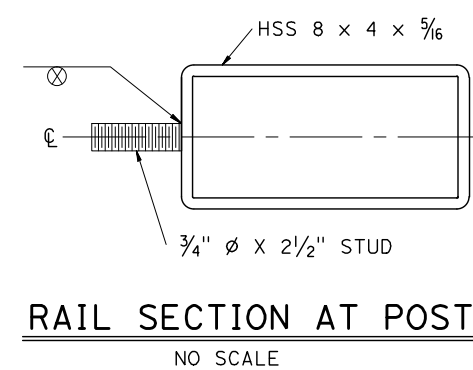
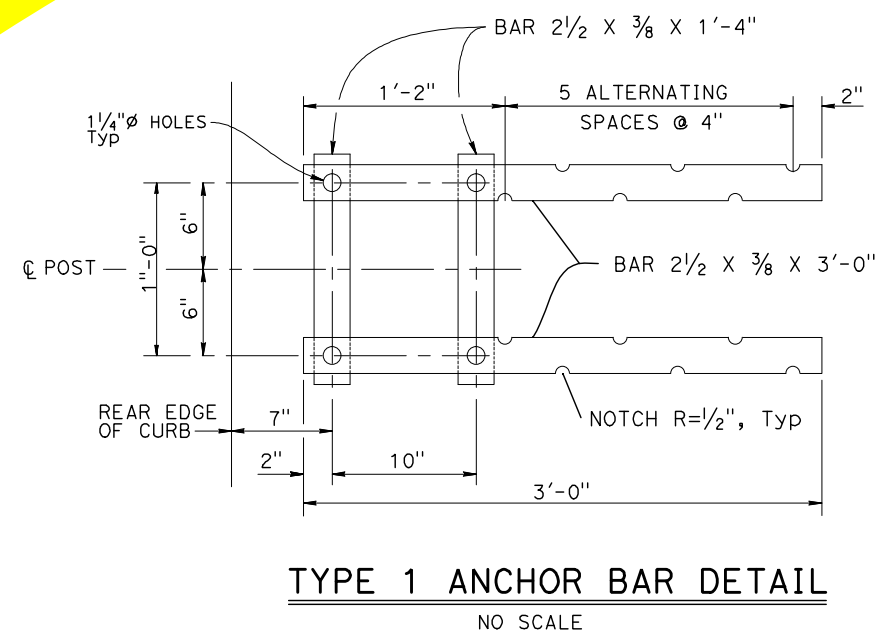
NOTE:
1. Anchor bolts may be tack welded to anchorage.



NOTE:
Shims as needed between posts and HSS rail tubes.



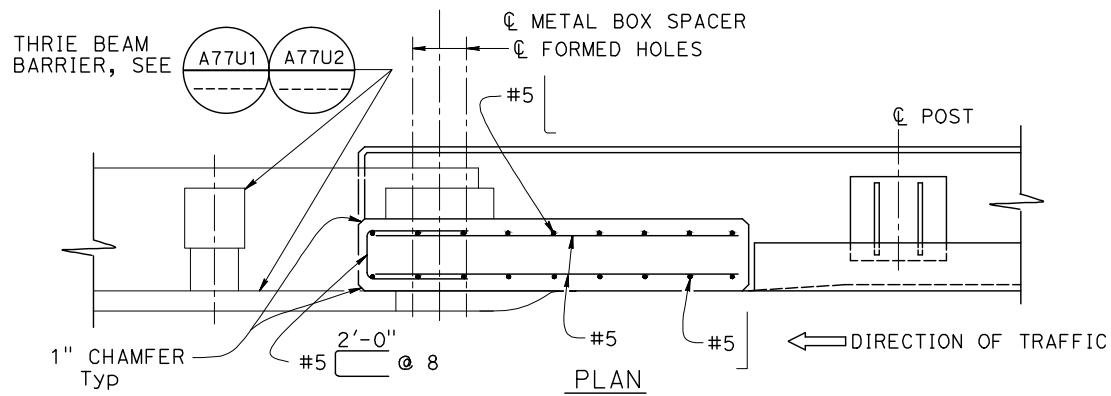
NOTE:
POST NOT SHOWN FOR CLARITY



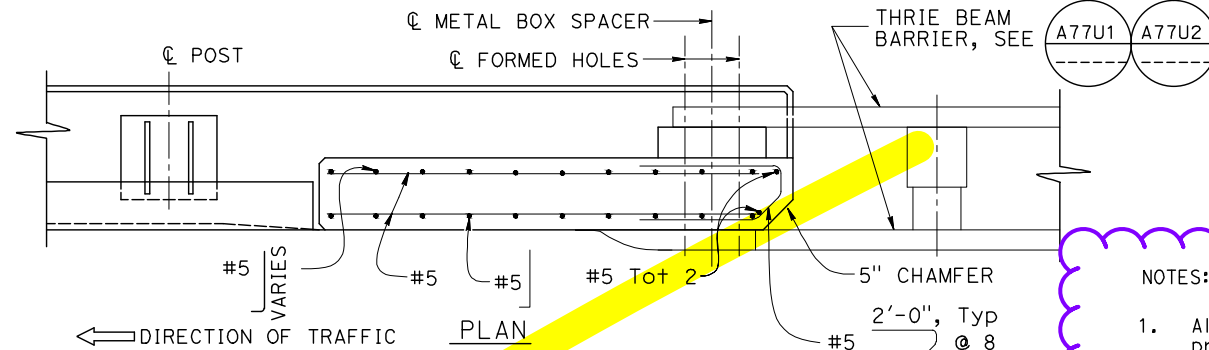
DESIGN OVERSIGHT	DESIGN	BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION		BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)		
SIGN OFF DATE	DETAILS	BY G. BOYKO	CHECKED T. PHAM	JASON HICKEY PROJECT ENGINEER		POST MILES	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.1		
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)	QUANTITIES	BY G. GUARDADO	CHECKED N. FELDE	UNIT: X PROJECT NUMBER & PHASE: X		DISREGARD PRINTS BEARING EARLIER REVISION DATES		REVISION DATES (PRELIMINARY STAGE ONLY)	
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DATE PLOTTED ==> \$DATE
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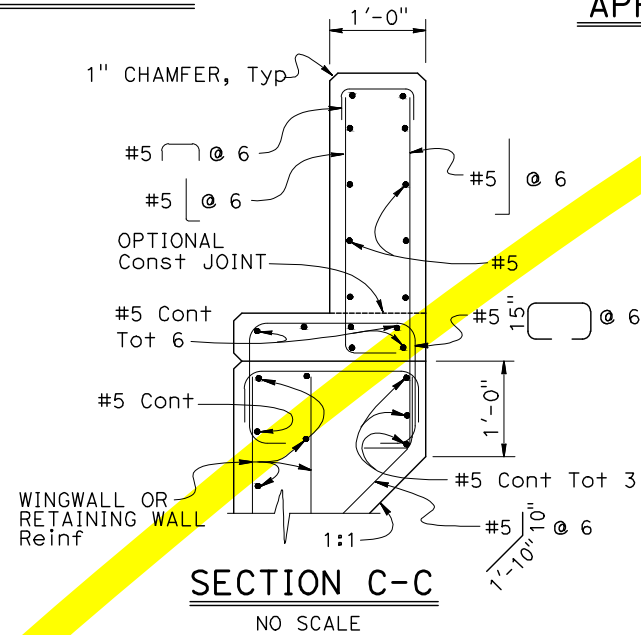
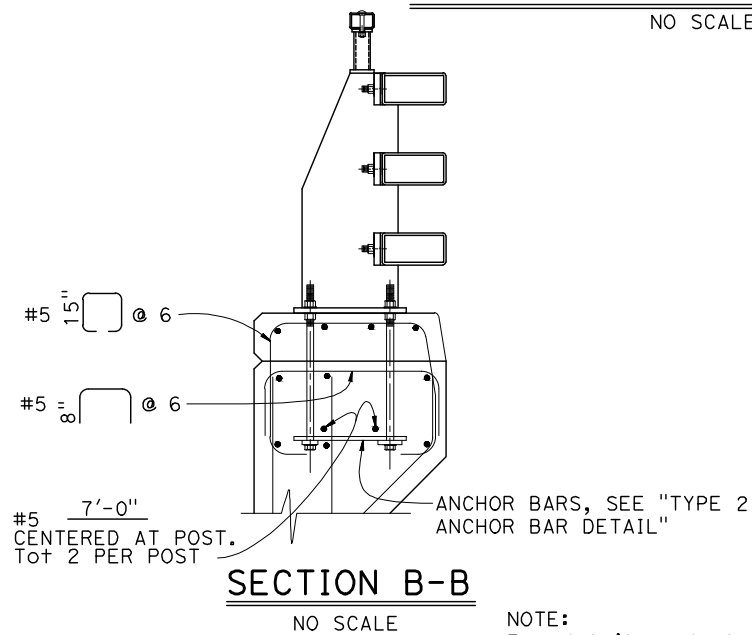
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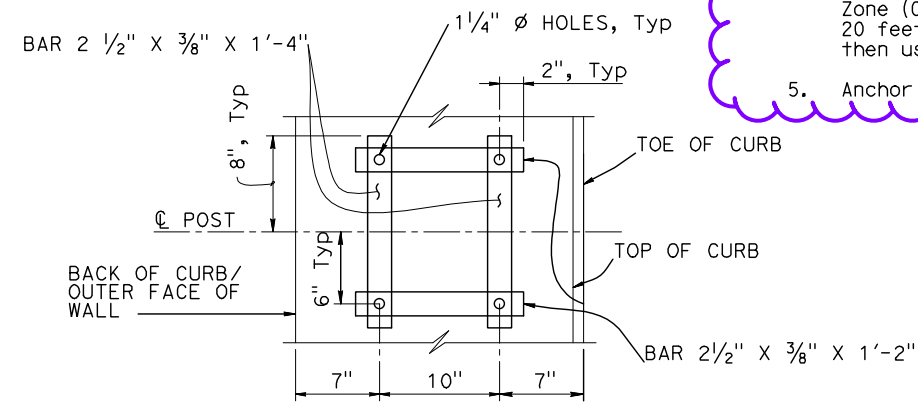
DEPARTURE END BLOCK DETAIL
NO SCALE



APPROACH TRANSITION BLOCK DETAIL
NO SCALE



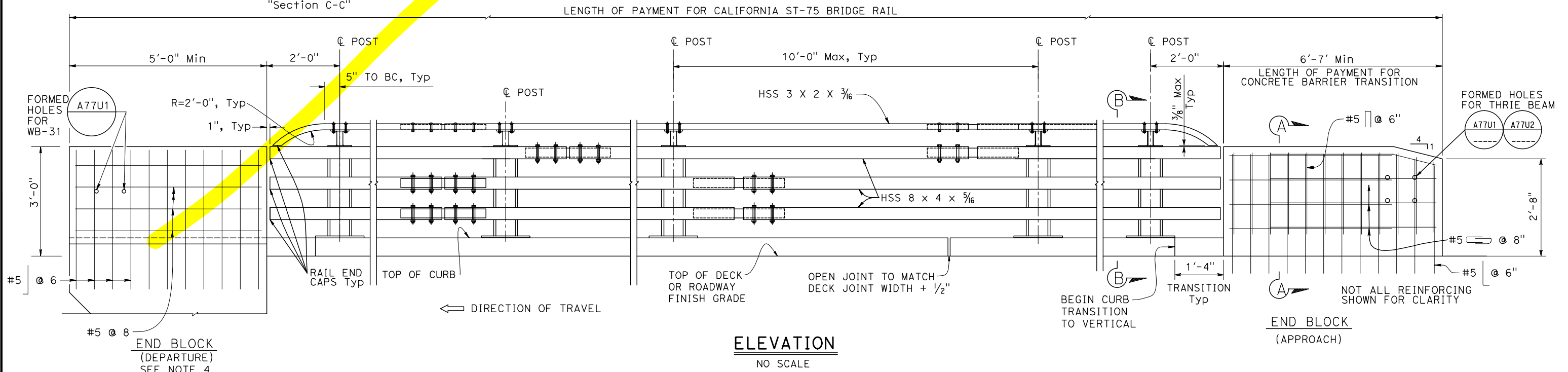
NOTE:
For details not shown see "Section C-C"



NOTES:

1. All horizontal and vertical profile grade to longitudinal profile grade.
2. Posts are normal to profile grade of structure.
3. Posts are vertical to the transverse cross section.
4. If departure end block is within the Clear Recovery Zone (CRZ, 30 feet for expressways and freeways and 20 feet for conventional highways) of opposing traffic, then use the approach end block at the departure end.
5. Anchor bolts may be tack welded to anchorage.

clean up



DESIGN OVERSIGHT

SIGN OFF DATE

DESIGN BY T. PHAM

DETAILS BY G. BOYKO

QUANTITIES BY G. GUARDADO

CHECKED Y. REN

CHECKED T. PHAM

CHECKED N. FELDE

PREPARED FOR THE
SACRAMENTO COUNTY
DEPARTMENT OF TRANSPORTATION

JASON HICKEY
PROJECT ENGINEER

BRIDGE NO.
24C0606

POST MILES

LAGUNA CREEK BRIDGE (REPLACE)
CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.2

DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)

ORIGINAL SCALE IN INCHES
FOR REDUCED PLANS

0 1 2 3

UNIT: X
PROJECT NUMBER & PHASE: X

DISREGARD PRINTS BEARING
EARLIER REVISION DATES

REVISION DATES (PRELIMINARY STAGE ONLY)

05/27/13 11/08/20

CONTRACT NO.: X

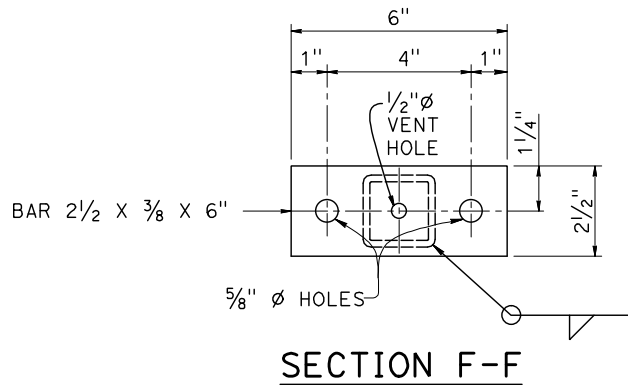
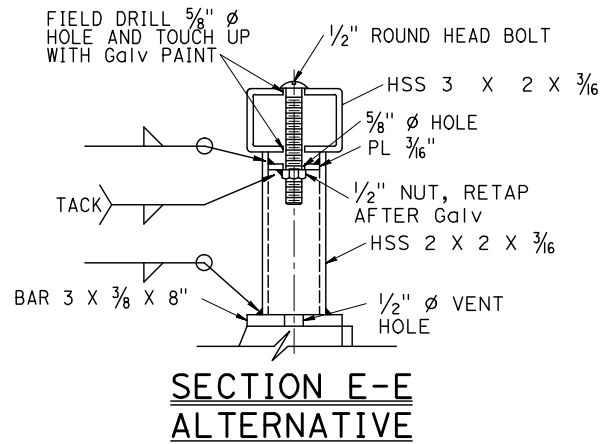
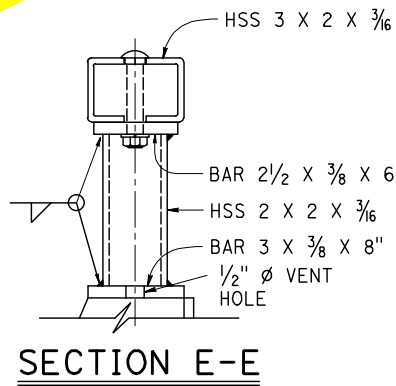
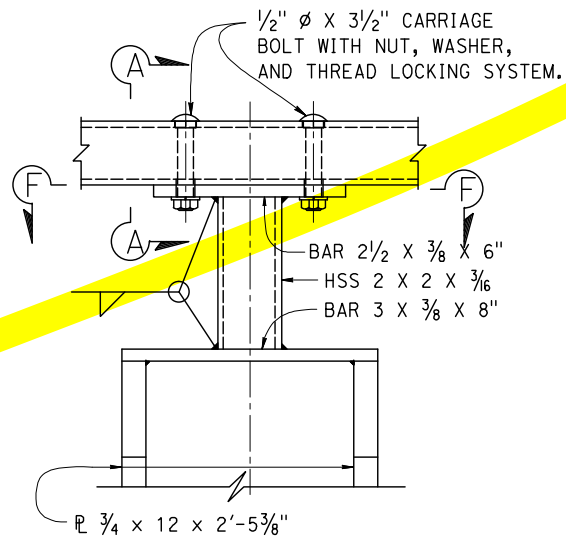
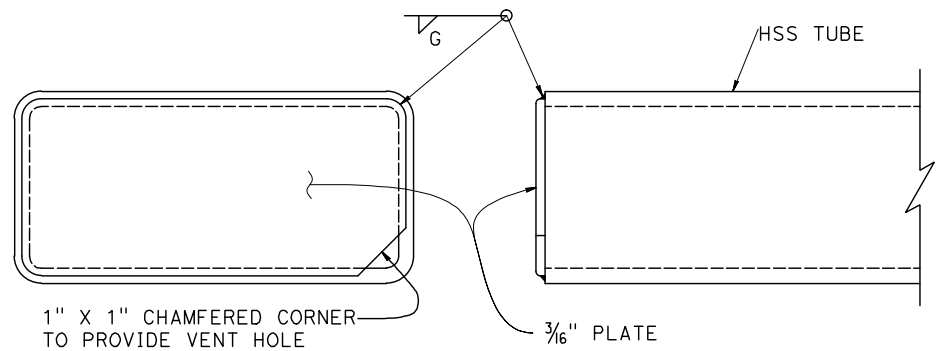
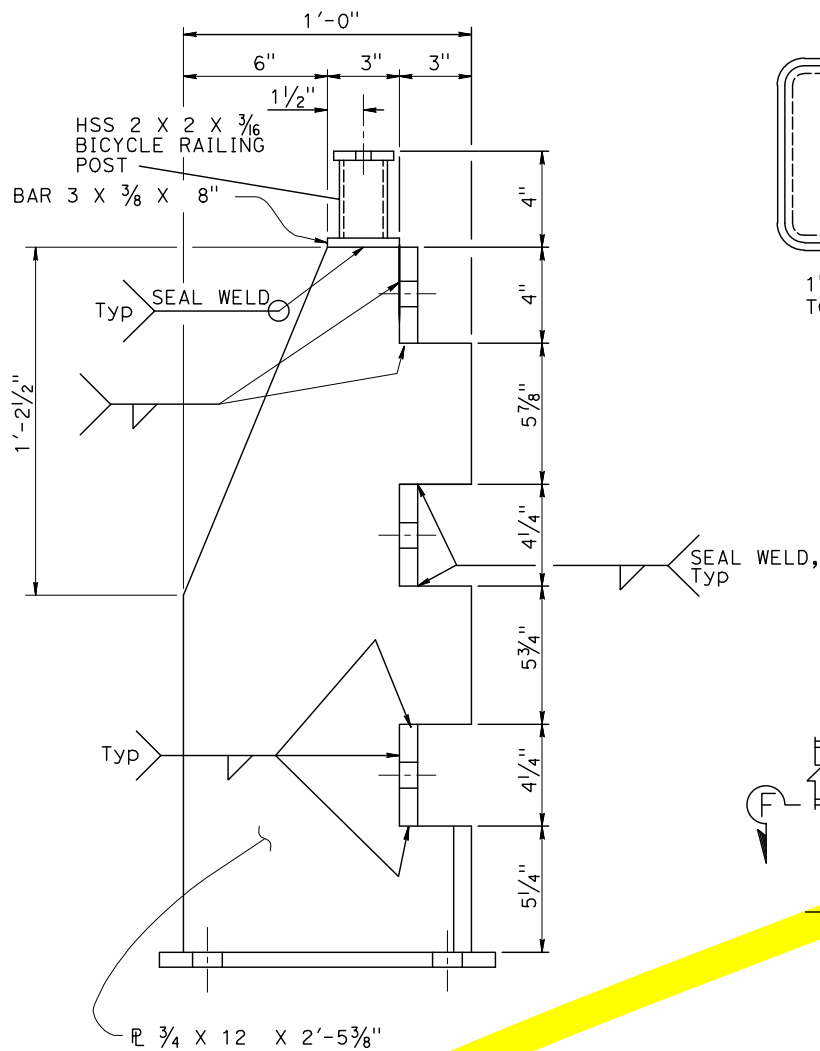
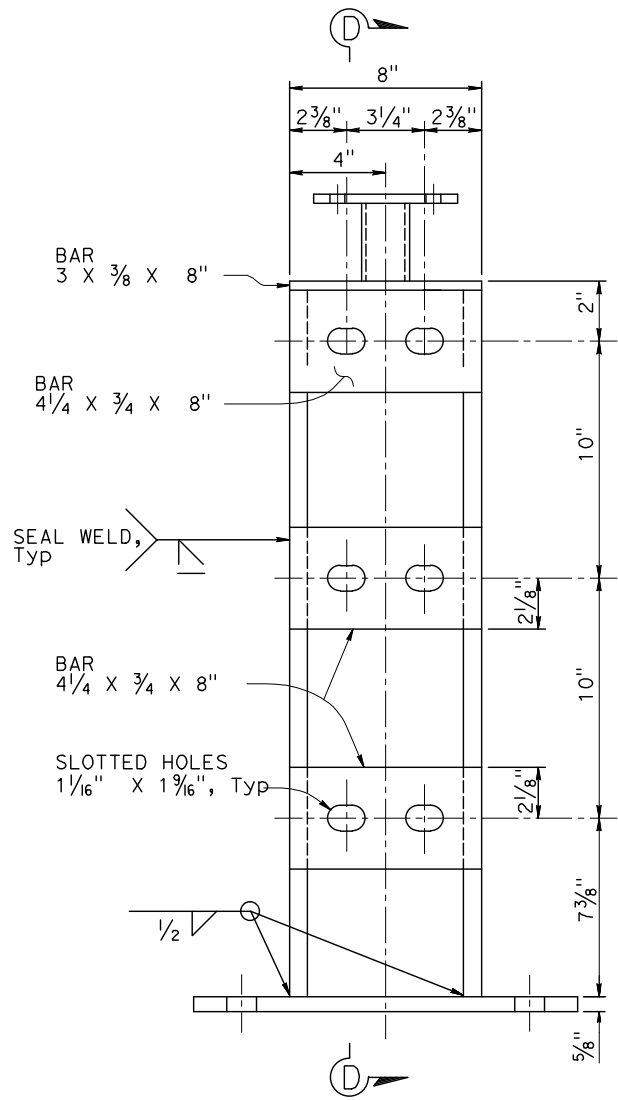
PROJECT ID:

SHEET 18 OF 23

DATE PLOTTED => \$DATE USERNAME => \$USER TIME PLOTTED => \$TIME

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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			DATE	11/6/20	
PLANS APPROVAL DATE			11/6/20		
The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.			REGISTERED PROFESSIONAL ENGINEER No. 81809 Exp. 3/31/22 CIVIL STATE OF CALIFORNIA		
SACRAMENTO COUNTY 411 BRANCH CENTER ROAD SACRAMENTO, CA 95827					
MARK THOMAS & COMPANY, INC. 7300 FOLSOM BOULEVARD, SUITE 203 SACRAMENTO, CA 95826					



BICYCLE RAILING CONNECTION DETAILS
NO SCALE

DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)	
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM			POST MILES	CALIFORNIA ST-75 BRIDGE RAIL DETAILS No.3	
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE	ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)	SHEET 19 OF 23
				FILE => \$REQUEST	CONTRACT NO.: X		PROJECT ID:

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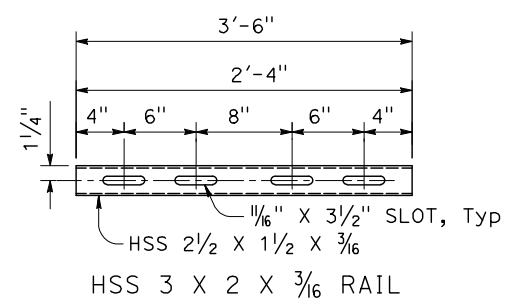
DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				
REGISTERED CIVIL ENGINEER			DATE	11/6/20	
PLANS APPROVAL			DATE		
The State of California or its officers or agents shall not be responsible for the accuracy or completeness of scanned copies of this plan sheet.			REGISTERED PROFESSIONAL ENGINEER		
No. 81809			Exp. 3/31/22		
CIVIL			STATE OF CALIFORNIA		

TRUONG VIET PHAM
No. 81809
Exp. 3/31/22
CIVIL
STATE OF CALIFORNIA

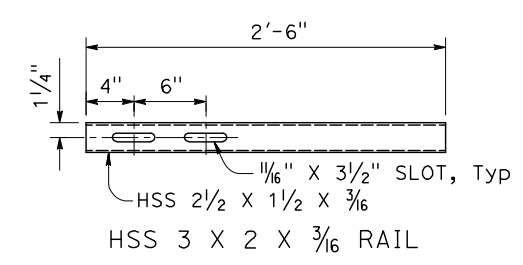
SACRAMENTO COUNTY
411 BRANCH CENTER ROAD
SACRAMENTO, CA 95827

MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826

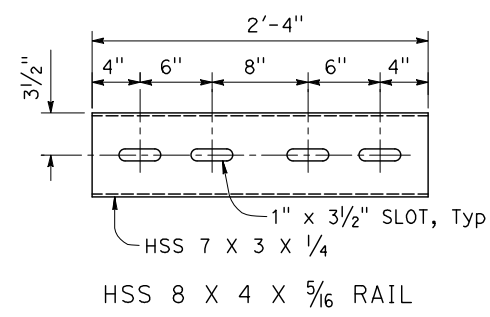
- NOTES:
1. HS bolts, nuts and washers, snug tightened, and thread locking system.
 2. Use 1/2"Ø X 3 5/16 BOLTS (HSS 3 X 2 X 3/16)
Use 3/4"Ø X 5 5/16 BOLTS (HSS 8 X 4 X 5/16)
 3. Each rail length must be continuous over a minimum of two posts.
 4. The fabricator must check that the tubular sleeve splices conform to the dimensions indicated to assure proper clearance.
 5. Except for expansion splices, not more than one splice permitted per same side of post.



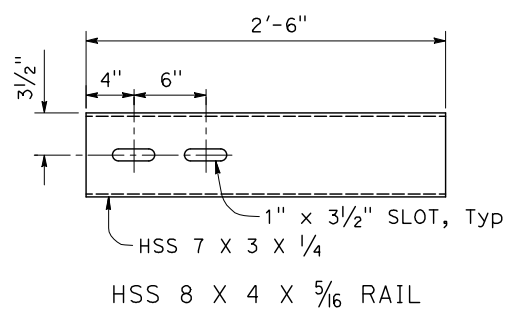
**STANDARD SLEEVES
DETAILS**
NO SCALE



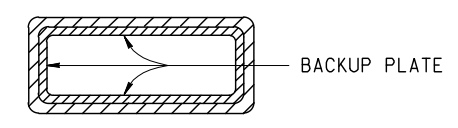
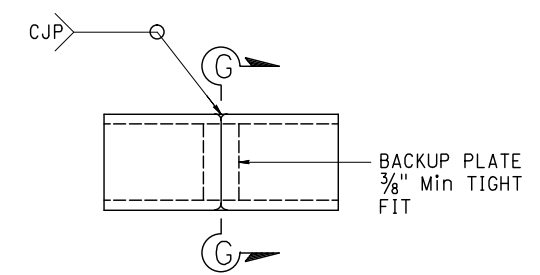
**EXPANSION SLEEVES
DETAILS**
NO SCALE



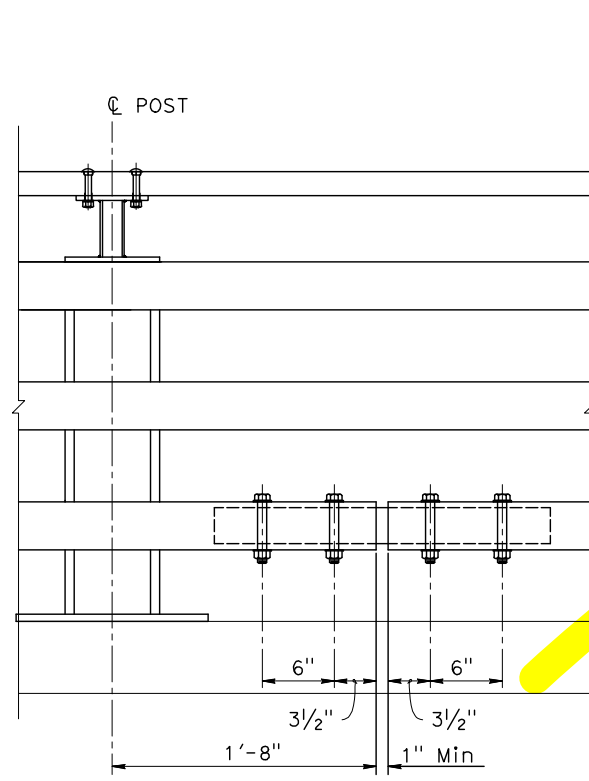
**STANDARD SLEEVES
DETAILS**
NO SCALE



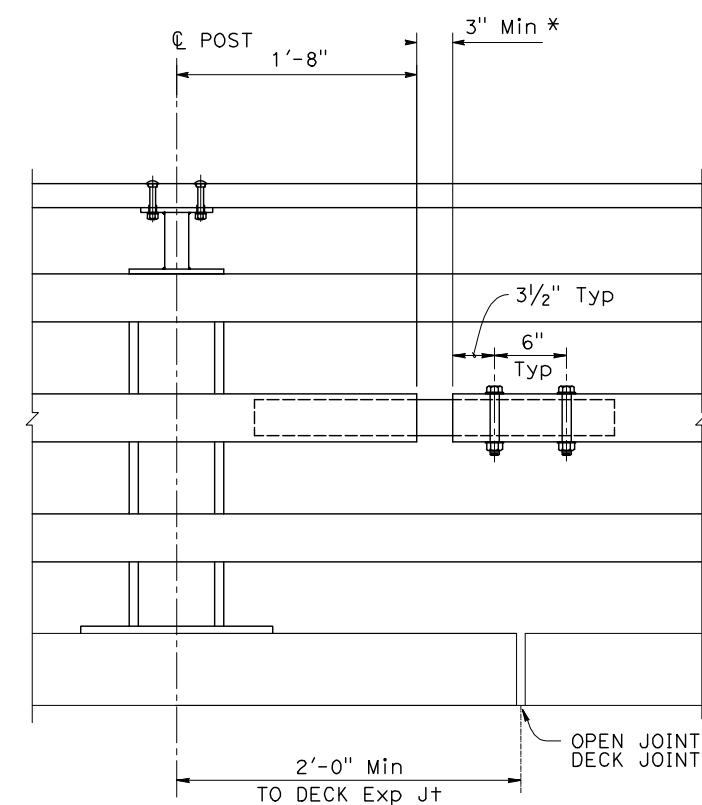
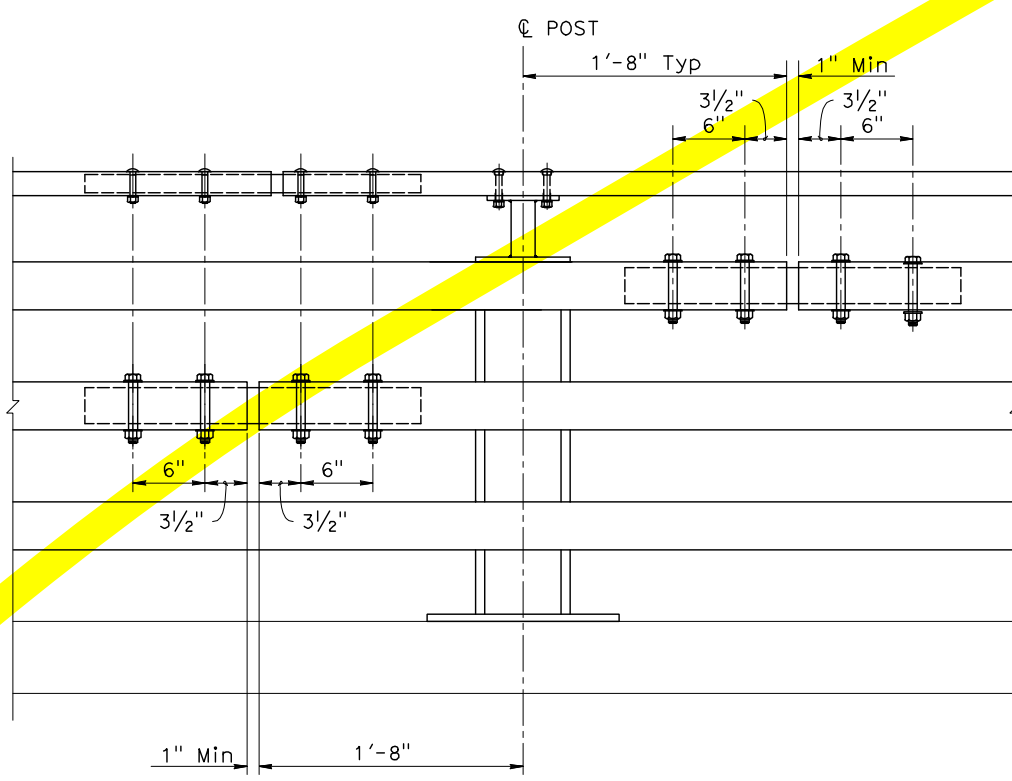
**EXPANSION SLEEVES
DETAILS**
NO SCALE



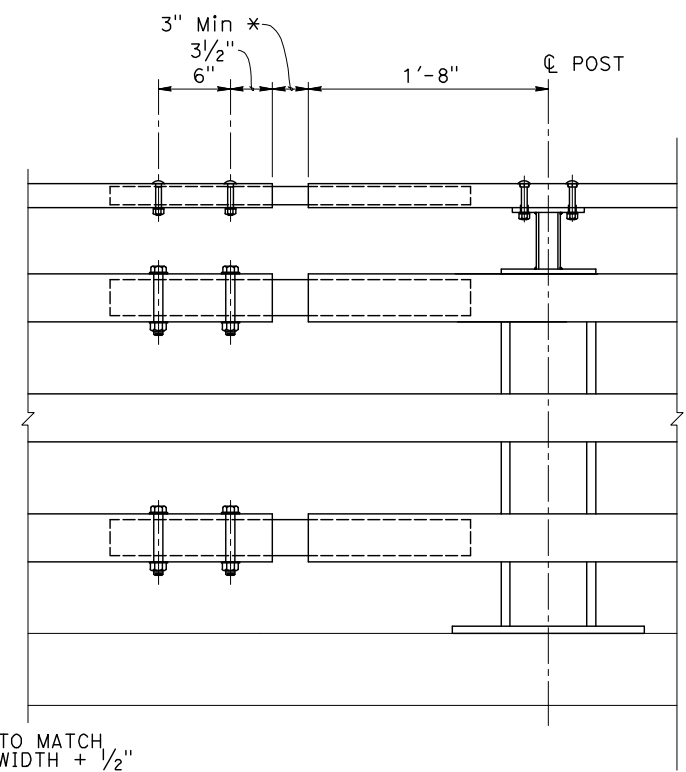
SECTION G-G
**ALTERNATE TUBE
WELDED STANDARD SPLICE**
NO SCALE



STANDARD SPLICE
NO SCALE



EXPANSION SPLICE
NO SCALE
* MATCH DECK OR WALL JOINT



DESIGN OVERSIGHT	DESIGN BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO. 24C0606	LAGUNA CREEK BRIDGE (REPLACE)	
SIGN OFF DATE	DETAILS BY G. BOYKO	CHECKED T. PHAM	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)	SHEET 20	OF 23
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)	QUANTITIES BY G. GUARDADO	CHECKED N. FELDE	ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	FILE ==> \$REQUEST	CONTRACT NO.: X	PROJECT ID:	

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DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				


REGISTERED CIVIL ENGINEER

11/6/20
DATE

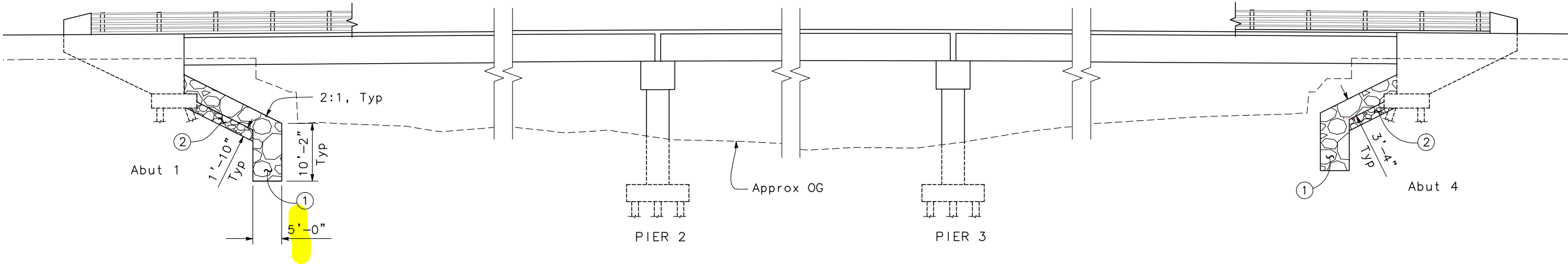
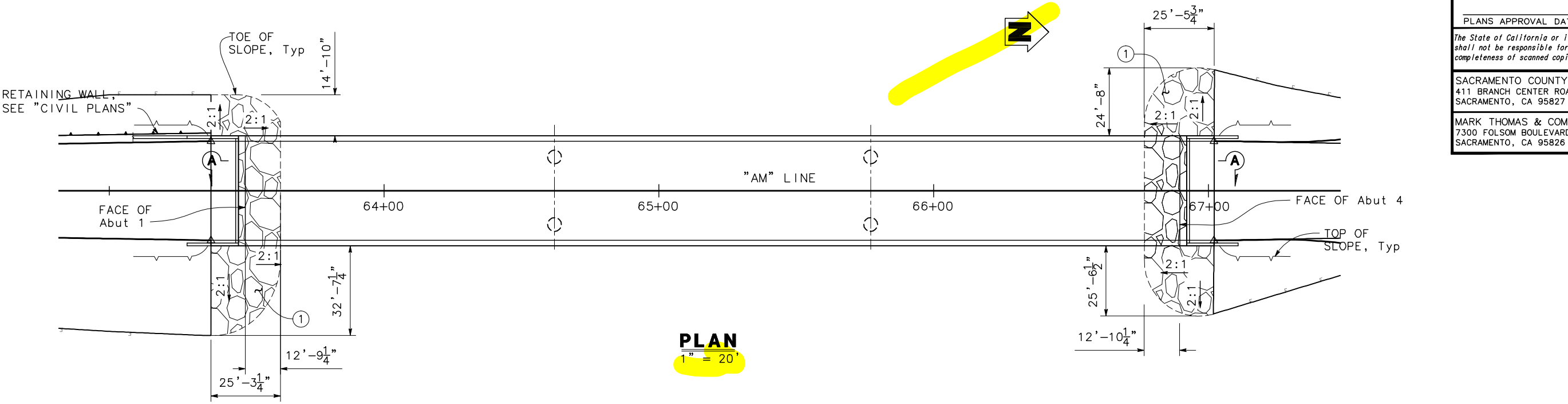
PLANS APPROVAL DATE

TRUONG VIET PHAM
No. 81809
Exp. 3/31/22
CIVIL
STATE OF CALIFORNIA

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SACRAMENTO COUNTY
411 BRANCH CENTER ROAD
SACRAMENTO, CA 95827

MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826



- LEGEND:**
- ① ROCK SLOPE PROTECTION (1/4 TON, METHOD B)
 - ② ROCK SLOPE PROTECTION (BACKING No. 1, METHOD B)

NOTE:
THE CONTRACTOR SHALL VERIFY ALL
CONTROLLING FIELD DIMENSIONS BEFORE
ORDERING OR FABRICATING ANY MATERIAL.

DESIGN OVERSIGHT	SIGN OFF DATE	DESIGN	BY T. PHAM	CHECKED Y. REN	PREPARED FOR THE SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION	JASON HICKEY PROJECT ENGINEER	BRIDGE NO.	24C0606	LAGUNA CREEK BRIDGE (REPLACE) ROCK SLOPE PROTECTION		
		DETAILS	BY G. BOYKO	CHECKED T. PHAM			POST MILES				
		QUANTITIES	BY G. GUARDADO	CHECKED N. FELDE							
DESIGN DETAIL SHEET (ENGLISH) (REV.7/16/10)					ORIGINAL SCALE IN INCHES FOR REDUCED PLANS	0 1 2 3	UNIT: X PROJECT NUMBER & PHASE: X	DISREGARD PRINTS BEARING EARLIER REVISION DATES	REVISION DATES (PRELIMINARY STAGE ONLY)	SHEET	OF
							FILE ==> \$REQUEST	09/08/18 04/28/14 12/02/14 05/18/15 08/11/16 09/15/16	CONTRACT NO.: X PROJECT ID:		

TIME PLOTTED ==> \$TIME
DATE PLOTTED ==> \$DATE
USERNAME ==> \$USER